Delayed and Deterred: Tuition Fees, University Participation, and the Selectivity of Entrants

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Abstract

How do students respond when university fees nearly triple, but nothing is paid upfront? I examine England's 2012 tuition fee reform using administrative data and a difference-in-discontinuities design exploiting the September school-entry cutoff. On-time university entry fell by 4.6 percentage points, with delayed entry offsetting roughly half the decline. Five-year participation remained 2.2 points lower. Among entrants, attendance at above-median and top-quartile institutions rose by 6.0 and 5.3 points, respectively, but unconditional gains in entry to selective universities across the full cohort were smaller. These patterns persist after accounting for concurrent changes to student number controls. The apparent shift towards more selective universities therefore largely reflects selection into higher education rather than widespread upgrading.

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1 Introduction

The cost of higher education plays a central role in human capital investment, shaping both individual decisions to pursue further study and governments' willingness to subsidise that investment. Classic human capital models predict that individuals invest in higher education when the present value of future returns, net of costs, exceeds the value of entering the labour market directly (Becker 1964). An increase in the price of university should therefore reduce investment at the margin. But a price change can operate on several margins at once. Students may respond by changing *whether* they enrol, *when* they enrol, or *where*. And because the students who continue into higher education after a price increase are a selected group, changes in the characteristics of entrants need not describe changes in the cohort from which they are drawn. Distinguishing these margins matters for policy: a fee increase that delays entry has different consequences from one that prevents it, and a change in the composition of entrants has different consequences from a change in the opportunities taken up across a cohort.

A widely cited rationale for subsidising higher education is the presence of credit constraints: individuals with strong academic potential but no collateral may be unable to finance tuition even when expected returns are high (Black et al. 2023; Lochner and Monge-Naranjo 2011). England's 2012 tuition fee reform provides a setting in which this friction is largely absent. The reform nearly tripled the cap on annual tuition fees, from £3,375 to £9,000, but all students retained access to government-backed income-contingent loans: no student faced an upfront payment, and repayment remained contingent on future earnings. The fee increase therefore operated through the expected cost of a degree and the debt students would accumulate, not through liquidity.

I estimate the effects of the reform using population-wide administrative data from the Longitudinal Education Outcomes (LEO) dataset, which links school and university records for entire cohorts, and a difference-in-discontinuities design built on England's 1st September school-entry cutoff. Children born on or after 1 September start school a year later than those born days earlier,

generating a well-documented discontinuity in educational outcomes at the cutoff. Because fees followed the year of university entry, individuals born in September 1993 were the first whose on-time entry fell under the new regime, while those born in August 1993 could enter on time under the old one. I compare the discontinuity at this cutoff with a pooled discontinuity estimated from the four preceding cutoffs, which provides the counterfactual: the discontinuity that would have been expected at the 2012 cutoff without the reform. Month of birth takes only a few values, so I estimate the underlying discontinuities with the framework developed for regression discontinuity designs with a discrete running variable (Kolesár and Rothe 2018; Armstrong and Kolesár 2020), and test the difference between the reform-year and pre-reform discontinuities with conventional standard errors.

A central difficulty is that the final pre-reform cohort was not entirely untreated. The reform was announced almost two years before it took effect, and the fee a student paid depended on when they entered university, not when they were born. Members of the 2011 entry cohort who took a gap year entered under the new fees, and the announcement gave others an incentive to change their plans. Because the 2011 cohort supplies the comparison group immediately below the 2012 cutoff as well as part of the pre-reform benchmark, its exposure biases the estimated effects on delayed and cumulative entry toward zero. I address this by modelling university entry in earlier cohorts as a function of students' characteristics, prior attainment, and school and neighbourhood fixed effects, and using the model to construct counterfactual outcomes for the 2011 cohort. For outcomes defined only among university entrants, such as the selectivity of the institution attended, I instead reweight the 2011 entrants so that their composition matches what the earlier selection process would have produced. The correction matters for the conclusions, and I report corrected and uncorrected estimates side by side.

Three findings emerge. First, the reform produced a large immediate contraction in university participation, about half of which persisted. On-time entry fell by 4.6 percentage points. Entry within one year of the on-time date was 2.5 points lower, and the gap then remained close to

2.2 points through five years, by which point the affected individuals were 23. About half of the immediate response was therefore postponement, almost all of it reversed within a year, and about half was not reversed within five years. Without the correction, the five-year estimate is about half as large and not statistically significant at the 5 percent level, the attenuation that contamination of the comparison cohort would produce.

Second, the reform changed where entrants enrolled. Conditional on entering university, students became 4.4 percentage points more likely to attend an above-median institution and 3.7 points more likely to attend a top-quartile institution; after reweighting the 2011 entrants, the estimates are 6.0 and 5.3 points. The response is not confined to students covered by the concurrent exemption of high-attaining students from student number controls: among students without AAB grades at A-level, the conditional estimates are 5.2 and 4.4 points.

Third, these conditional changes overstate how far the cohort as a whole moved up the selectivity distribution. Measured across the entire cohort, the probability of attending a top-quartile institution rose by 0.9 percentage points after correction, while the effect on above-median attendance, 0.5 points, is small and not significant at the 5 percent level. Together with the fall in participation, the point estimates imply that the contraction was concentrated among institutions below the median, and that much of the improvement in the selectivity of entrants reflects who continued into higher education. Subject choice shows a related pattern: the share of the cohort studying STEM subjects fell by 1.5 points after correction, while the STEM share among entrants did not change detectably. The estimates for degree completion, withdrawal for poor academic performance and progression to postgraduate study are not statistically significant.

These results contribute to three literatures. The first studies how tuition prices affect participation. Evaluations of the 2012 reform have generally concluded that it had little effect on overall enrolment or on the socioeconomic composition of entrants (Murphy et al. 2019; Azmat and Simion 2021). The design used here differs in two respects that matter for comparing the evidence: it isolates the cohort boundary at which the fee regime changed, differencing out the

secular growth in participation over the period, and it follows each individual's entry for five years rather than measuring a single entry year. It reveals a sizeable immediate response and a contraction in participation of about 2.2 points that had not been reversed five years later. The second literature concerns where students enrol. I show that the selectivity of entrants rose substantially, but that the cohort-level change was much smaller, so that conclusions about the destination margin depend on whether outcomes are measured among entrants or across the cohort. The third concerns the evaluation of pre-announced policies. When treatment depends on the timing of a choice, nominally untreated cohorts can be partially treated (Malani and Reif 2015). I show how such contamination enters a difference-in-discontinuities design built on adjacent cohorts, why it falls with particular weight on the cohort immediately preceding the reform, and how it can be corrected for outcomes defined over the whole cohort and over a selected sample.

The policy implications are specific to England's institutional setting. Income-contingent loans removed upfront costs, but they did not prevent a large immediate response, nor a persistent contraction in participation among the individuals studied here. At the same time, the reform coincided with a reallocation of entrants toward more selective institutions, which may reflect closer attention to expected returns, changes in who chose to enter, or responses by institutions. I do not evaluate the welfare consequences of these changes, which depend on the returns to the education forgone and to the reallocation, and which this design does not identify.

The remainder of the paper is structured as follows. Section 2 reviews the related literature. Section 3 provides institutional background on the English higher education system and the 2012 reform. Section 4 describes the data. Section 5 sets out the empirical strategy, including the correction for the final pre-reform cohort. Section 6 presents the results, Section 7 discusses mechanisms and interpretation, Section 8 discusses robustness and limitations, and Section 9 concludes.

2 Related Literature

Human capital theory predicts that students weigh the costs of university against its expected long-term returns, and that an increase in price should reduce investment at the margin (Becker 1964; Mincer 1974). This prediction concerns the *level* of investment, and empirical work on tuition prices has largely inherited that focus, asking whether fees reduce participation. A price change can also operate on margins that a participation regression cannot see: the timing of entry, particularly when the change is announced in advance, and the choice of institution among those who enter. This paper measures all three margins within a single design, and shows that conclusions about the destination margin depend on whether it is measured among entrants or across the cohort.

A large literature examines the participation margin directly. In North America, where tuition varies across states, provinces, and institutions, enrolment falls modestly when sticker prices rise (Kane 1995; Neill 2009; Hemelt and Marcotte 2011); see Deming and Dynarski (2009) for a review. The closest international analogue to the English experience is Germany, where the staggered introduction of tuition fees across states reduced enrolment intentions and transitions into university by small amounts (Hübner 2012; Bruckmeier and Wigger 2014). These studies typically measure enrolment in a fixed window around the price change, and therefore cannot distinguish delay from a longer-lasting reduction in participation.

A related American literature studies financial aid rather than posted prices, finding that need-based grants raise enrolment and persistence among low-income students (Dynarski 2003; Bettinger 2004; Castleman and Long 2016), with downstream benefits for completion and earnings that can exceed their fiscal cost (Denning et al. 2019; Black et al. 2023). This work is conceptually distinct: aid relaxes liquidity constraints for students who cannot borrow against future earnings, whereas a fee change under a universal income-contingent loan system alters the expected cost of a degree without any liquidity effect. Because no student faces an upfront payment and re-

payment is contingent on earnings, debt aversion (Callender and Jackson 2005; Field 2009) and responses to the headline price are the main behavioural channels through which such a reform could deter entry.

The UK evidence on tuition fees is mixed. Earlier fee increases reduced participation modestly (Dearden et al. 2011; Sá 2014), and evaluations of the 2012 reform conclude that it had little effect on overall enrolment or on the socioeconomic composition of entrants (Murphy et al. 2019; Azmat and Simion 2021), with Azmat and Simion (2021) additionally reporting minimal effects on early labour market outcomes. Yet the raw entry data show a sharp drop in 2012 (Appendix Figure A.2). My estimates indicate that this drop was not merely a matter of timing: about half of the on-time response persisted for at least five years. Because existing studies rely on application-year snapshots or survey data, they also observe neither entry at later ages nor where a cohort ultimately enrolled; linked administrative data that record entry at any age allow me to measure both.

The paper also relates to work on pre-announced policies. When agents can shift the timing of a decision, comparisons of outcomes immediately before and after implementation confound the policy's effect with intertemporal substitution around it, and pre-period observations can be contaminated as controls (Malani and Reif 2015). The 2012 reform, announced almost two years before implementation with fees tied to the year of entry, created exactly these conditions. Rather than assume that the final pre-reform cohort provides a clean comparison, I correct its outcomes using the behaviour of earlier cohorts.

Methodologically, the design combines a regression discontinuity with a before–after comparison, a difference-in-discontinuities approach used in other policy settings (Grembi et al. 2016). The running variable, month of birth, is discrete. With a discrete running variable the discontinuity at the cutoff must be extrapolated from a few support points, and the resulting specification error does not vanish in large samples. Lee and Card (2008) proposed clustering standard errors by values of the running variable to account for this error; Kolesár and Rothe

(2018) show that such confidence intervals can perform poorly and propose alternatives with guaranteed coverage under explicit restrictions on the conditional expectation function, building on the bias-aware intervals of Armstrong and Kolesár (2020). Section 5.3 explains which parts of this framework the paper uses.

The identifying variation rests on the relative age effect, the well-documented advantage of autumn-born children within academic cohorts (Russell and Startup 1986; McPhillips and Jordan-Black 2009; Crawford et al. 2013, 2014; Black et al. 2011). Season-of-birth selection, the classic threat to birthdate designs (Buckles and Hungerman 2013), is differenced out to the extent that it is stable across cohorts, and I report balance tests on predetermined characteristics for each cohort in the sample.

3 Institutional Background

In the decades following WWII, the funding of students into higher education was largely administered by local authorities, an approach formalised by the Education Act (1962). The Act mandated that Local Education Authorities pay tuition fees and provide means-tested maintenance grants for eligible full-time university students (UK Parliament 1962), making undergraduate education effectively free at the point of use across the UK. The economic rationale was summarised in the Robbins Report of 1963, which recommended a substantial expansion of the university system on the grounds that higher education generated considerable positive social and financial externalities, such that growth in the economy and in tax receipts would deliver a positive return on public investment (Committee on Higher Education 1963).

This assumption came under strain in the 1990s. Student numbers had grown rapidly while funding per pupil declined by 40% between the early 1980s and 1997, putting considerable pressure on university budgets (Dearden et al. 2011). At the same time, the case for higher education paying for itself at the exchequer level had begun to wane: the Dearing Report (1997) argued that

a cost-sharing approach would be equitable since graduates were the primary beneficiaries of university education (National Committee of Inquiry into Higher Education 1997). Subsequent evidence supports this reassessment: Britton et al. (2020) estimate that while average lifetime returns to the exchequer are positive net of subsidies, these returns are driven predominantly by the highest earners, and total returns are negative for up to 30% of graduates.

3.1 Post-1998 Reforms

Following the Dearing Report, the Teaching and Higher Education Act (1998) introduced annual tuition fees of £1,000, paid upfront by students, while maintenance grants were replaced with income-contingent maintenance loans (UK-Government 1998; Hillman 2013). Exemptions applied to students from low-income families, and following devolution in 1999 the UK nations began to diverge: Scotland introduced a graduate endowment in lieu of upfront fees and by 2007 had abolished tuition fees entirely for Scottish-domiciled undergraduates at Scottish institutions (Sá 2014).

By the early 2000s the £1,000 cap was yielding limited revenue, and the requirement that fees be paid upfront was widely seen as a barrier to access. The Higher Education Act 2004 raised the cap to £3,000 per year from the 2006–07 academic year and, crucially, ended upfront payment altogether (UK Parliament 2004). Fees would instead be covered by government-backed loans, repaid after graduation at 9% of earnings above a £15,000 threshold, with any outstanding balance written off after 25 years. University thereby became free at the point of use once more, with higher fees operating as a reduction in expected lifetime income rather than a liquidity burden. This institutional detail distinguishes the UK setting from the US, where the policy debate centres on the under-allocation of capital to students who cannot collateralise their future earnings, and it is what allows this paper to interpret responses to the 2012 reform as responses to price rather than to credit constraints.

3.2 The 2012 Reform and Its Announcement

By 2010, recession and the declining real value of the capped fee had renewed the pressure on university finances. The independent Browne Review, published in October 2010 recommended a market-driven abolition of the fee cap, effective for students entering university from the 2012–13 academic year (Independent Review of Higher Education Funding and Student Finance 2010). This was accepted by the government in November, but with the price cap rising to £9,000 rather than being abolished. In theory variable fees were intended to generate price competition; in practice almost all universities charged the maximum immediately upon implementation (House of Commons Library 2018). The reform also shifted the repayment threshold from just below £16,000 to £21,000, extended the repayment period from 25 to 30 years, and introduced a higher, income-contingent interest rate (House of Commons Library 2018). The net effect was a more progressive schedule: the lowest-earning graduates could expect to pay less per month than under the previous system (Britton et al. 2017). Appendix Figure A.1 summarises the evolution of the fee cap, with the 2012 increase standing as a clear outlier against the earlier reforms.

Two features of the reform's implementation are central to the empirical design. First, the fee regime a student faced was determined by their year of entry, not their year of birth: anyone commencing study before September 2012 paid the old fees for the duration of their course, while anyone commencing from September 2012 onwards paid the new fees. Second, the reform was announced almost two years before it took effect. Together these features created incentives to retime entry around the reform. Students in the final pre-reform cohort could lock in the lower fees by entering in 2011 rather than deferring, but those who nevertheless deferred entered under the new fees; students in the first treated cohort faced the higher fees whenever they entered. Appendix Figure A.4 shows that the share of students delaying entry by one year fell in 2011 by just over two percentage points. The final pre-reform cohort was therefore partially exposed to the reform, which Section 5.5 shows has important consequences for the estimates and corrects.

Appendix Figure A.2 displays the raw trend in domestic undergraduate enrolment over the

sample period: steady year-on-year growth, a sharp contraction in 2012, and a recovery in subsequent years. Taken at face value, the 2012 dip suggests a substantial deterrent effect of the fee increase, but raw entry counts mix the response to the reform with trends in cohort size and participation and with shifts in the timing of entry. The task of this paper is to estimate how much of the change reflects a causal response to the fee increase, how much of that response was postponement, and how the reform changed where students enrolled.

3.3 Concurrent Policies

Three contemporaneous policy changes could in principle confound the analysis. First, and most importantly, the regulation of student numbers changed in 2012–13. Entry controls had limited the number of fundable full-time undergraduate entrants at each institution since 2009–10. For 2012–13 entry, students with qualifications equivalent to AAB or higher at A-level were removed from these controls, and up to 20,000 places were withdrawn from institutions' core allocations and made available to institutions charging an average net fee of £7,500 or less (Higher Education Funding Council for England 2011). The exemption was extended to ABB+ in 2013–14, and number controls were abolished entirely from 2015–16. The AAB exemption gave selective institutions freedom to expand recruitment of high-attaining students at the moment the fee reform took effect, and is therefore a candidate explanation for any shift in enrolment toward selective institutions. Section 6.4 examines whether the shift is present among students below the AAB threshold. Second, the participation age was raised, requiring young people in England to remain in education or training until age 17 from 2013 and 18 from 2015; this affects younger cohorts than those studied here. Third, maintenance support evolved over the period, but these changes were incremental and applied smoothly across cohorts (House of Commons Library 2018), so that they are differenced out to the extent that they did not change discontinuously at the 2012 cutoff.

4 Data and Descriptive Statistics

This paper uses data from the Longitudinal Education Outcomes (LEO) dataset. This administrative panel links school census records from the National Pupil Database (NPD) on student characteristics to Higher Education Statistics Agency (HESA) records of university enrolment and outcomes, matched further to HMRC administrative data on earnings, employment, and benefits. The result is a longitudinal panel covering over 30 million people across a 23-year period (1998–2020), from which I construct cohorts of English-domiciled, state-educated pupils defined by academic year of birth. Because the panel follows individuals rather than entry classes, I observe university entry at any age, not merely entry in the year following a student's 18th birthday.

Enrolment. For each individual I construct an indicator for on-time entry, defined as entry in the academic year following the individual's 18th birthday, and indicators for cumulative entry within one to five years of that date. Entry within five years corresponds to entry by age 23, and for the 2012 cohort requires HESA coverage through the 2017–18 academic year, which the panel provides.

Selectivity. Institution selectivity is measured by ranking institutions according to the mean A-level points of their entering students in the 2011 pre-reform reference year, from which I define indicators for attending an institution above the median, in the top quartile, in the top 20 percent, and in the top 10 percent of the distribution of percentiles of institutions. Each indicator is constructed in two forms. The *conditional* form is defined among individuals observed entering university and describes where entrants enrolled. The *unconditional* form is defined for the whole cohort and equals zero for individuals who do not enter, so that it measures the share of the cohort attending an institution of a given selectivity.

Subject choice, completion and progression. I also measure whether a student enrols in a STEM subject (conditionally and unconditionally), completes the degree, progresses to postgraduate study, or withdraws because of poor academic performance. The withdrawal indicator is constructed from the HESA reason for ending a course and isolates withdrawal for poor academic performance; it is not a general measure of non-completion. Conditional samples differ slightly across these outcomes.

Characteristics. Personal characteristics recorded in the NPD are used for balance tests, subgroup comparisons, and the correction described in Section 5.5. Family income is not recorded, so eligibility for free school meals (FSM) in year 11, a means-tested programme for children from low-income families, is used as a proxy for childhood poverty. Neighbourhood disadvantage is measured by school residence in a local area with low university participation (LOPAR). Prior attainment is measured by the number of GCSEs achieved at each grade, and by A-level results, from which I construct an indicator for achieving grades of AAB or higher, the threshold for the 2012–13 exemption from student number controls (Section 3.3). I also observe each student's school and Lower Layer Super Output Area (LSOA) of residence.[1]

Table 1 presents summary statistics for first-year university entrants immediately before and after the reform. The composition of entrants remained broadly stable between the 2011 and 2012 cohorts: the share of female students was unchanged, and the shares eligible for FSM and from low-participation areas rose marginally. Two margins moved. Entry in the year following students' 18th birthdays fell by 2.3 percentage points, and the share of entrants attending an above-average institution rose from 39 to 43 percent. Neither difference can be interpreted causally: raw comparisons of adjacent cohorts mix the reform with trends in participation and with the partial exposure of the 2011 cohort, which the design in Section 5 addresses.

[1] In accordance with the disclosure rules governing LEO, all reported counts are rounded to the nearest five.

Table 1: University Entrant Characteristics Before and After the 2012 Fee Reform

	2011	**2012**
Entry into university at 18 (%)	24.8	22.5
Female (%)	55.7	55.7
Eligible for FSM (%)	9.0	9.6
From low university-share area (%)	56.0	56.8
AAB or higher at A-level (%)	20.2	20.3
Attending above-average institution (%)	39	43

Notes: This table reports the share of first-year university entrants by key demographic and background characteristics in 2011 (pre-reform) and 2012 (post-reform). FSM = Free School Meals. The first row shows entry rate with the entire cohort as the denominator, while subsequent rows are conditional on entering HE. "Low university-share area" indicates local areas with below-average university participation in 2011. "Attending above-average institution" refers to institutions ranked above the mean in terms of average A-level points of entrants. Counts rounded to the nearest five per LEO disclosure rules.

5 Empirical Strategy

This paper asks whether a sharp increase in the cost of university affected students' decisions about whether, when, and where to attend. Estimating these effects is difficult: tuition fees typically change alongside other policies, participation was growing over the period, macroeconomic conditions shift the value of study relative to work, and, in this case, the reform's announcement changed the behaviour of the cohort immediately before it. A simple comparison of adjacent cohorts cannot separate the fee increase from these influences.

5.1 The School-Entry Discontinuity

In England, children are placed into academic cohorts by date of birth, with a cutoff on 1 September. Those born just before the cutoff (in August) enter school a full year earlier than those born just after (in September), even though they are only days apart in age. This generates sharp and well-documented differences in long-term outcomes, including university attendance, between children born on either side of the cutoff, commonly referred to as the relative age effect (Russell and Startup 1986; McPhillips and Jordan-Black 2009; Campbell 2013; Crawford et al. 2013, 2014). Appendix Figure A.3 shows this premium in university enrolment: September-born indi-

viduals are around four percentage points more likely to enrol than those born in August, with enrolment declining across subsequent birth months.

I index each cutoff by t, the academic year in which individuals born immediately after it would enter university on time. Individuals born in the months before cutoff t belong to entry cohort $t-1$ and are the youngest in their school year; those born in the months after it belong to entry cohort t and are the oldest in theirs. The reform cutoff is $t = 2012$: individuals born in September 1993 were the first whose on-time entry fell under the new fees, while those born in August 1993 belong to the 2011 entry cohort and could enter on time under the old regime. The pre-reform cutoffs are $t \in \{2008, 2009, 2010, 2011\}$.

Let x_i denote individual i's month of birth measured relative to the cutoff, and Y_i an outcome. The discontinuity at cutoff t is

$$\tau_t \;=\; \lim_{x \downarrow 0} \mathbb{E}\left[Y_i \mid x_i = x,\, t\right] \;-\; \lim_{x \uparrow 0} \mathbb{E}\left[Y_i \mid x_i = x,\, t\right]. \tag{1}$$

Because month of birth is discrete, the conditional expectation function is observed only at a small number of support points, and the limits in (1) are extrapolations from the points nearest the cutoff. Section 5.3 returns to the implications.

5.2 The Difference-in-Discontinuities Estimator

In every cohort, τ_t reflects the relative age effect together with any differences between the adjacent cohorts on either side of the cutoff. The pre-reform discontinuity therefore provides the counterfactual for 2012: had the reform not occurred, the discontinuity at the 2012 cutoff should have resembled those at earlier cutoffs. The main estimate is

$$\widehat{\Delta} \;=\; \widehat{\tau}_{2012} \;-\; \widehat{\tau}_{2008\text{–}2011}, \tag{2}$$

where $\widehat{\tau}_{2012}$ is the estimated discontinuity at the 2012 cutoff and $\widehat{\tau}_{2008\text{–}2011}$ is estimated from a single regression discontinuity fitted to the pooled observations from the four pre-reform cutoffs, with the running variable measured relative to each year's cutoff alongside cohort fixed effects. The pooled estimate uses all pre-reform observations in one fit; it need not equal the arithmetic average of four separately estimated annual discontinuities.

Let θ denote the effect of the reform on individuals at the 2012 cutoff, and τ^0_{2012} the discontinuity that would have been observed at that cutoff without the reform. $\widehat{\Delta}$ estimates θ if $\tau^0_{2012} = \tau_{2008\text{–}2011}$: absent the reform, the discontinuity at the 2012 cutoff would have equalled the pooled pre-reform discontinuity.

To see what this assumption requires, it is useful to decompose each discontinuity. Suppose, for exposition, that the expected outcome of an individual in entry cohort k with relative birth position x can be written as $m(x) + \mu_k$, where m is a relative-age profile, common to all cohorts, that jumps by ρ at the cutoff, and μ_k is a shift common to all members of cohort k. Then $\tau_t = \rho + \mu_t - \mu_{t-1}$. If the pooled estimator places weight ω_t on cutoff t, with $\sum_t \omega_t = 1$, the estimand of (2) is

$$\Delta \;=\; \theta \;+\; (\mu_{2012} - \mu_{2011}) \;-\; \sum_{t=2008}^{2011} \omega_t \,(\mu_t - \mu_{t-1})\,. \tag{3}$$

Three implications follow. First, the relative age effect ρ cancels, as does any season-of-birth selection that is stable across cohorts. Second, when the weights are approximately equal, cohort-level shifts that follow a linear trend also cancel, because the change between the 2011 and 2012 cohorts then equals the average annual change over the pre-reform period; the design therefore accommodates the secular growth in participation over this period. Third, a shift specific to the 2011 cohort enters Δ with coefficient $-(1 + \omega_{2011})$, because that cohort supplies both the comparison group immediately below the 2012 cutoff and the group immediately above the 2011 cutoff. Any effect of the reform on the 2011 cohort is therefore transmitted more than one-for-one

into the estimate. Section 5.5 addresses this.

Inference on Δ uses the standard errors of the two estimated discontinuities and assumes that their covariance is zero:

$$SE\big(\widehat{\Delta}\big) \;=\; \sqrt{SE\big(\widehat{\tau}_{2012}\big)^2 \;+\; SE\big(\widehat{\tau}_{2008\text{–}2011}\big)^2}. \tag{4}$$

I test $H_0 : \Delta = 0$ with the conventional two-sided test based on $\widehat{\Delta}/SE(\widehat{\Delta})$ and standard normal critical values. All p-values and significance levels reported for the main estimates are from this test. The zero-covariance assumption holds exactly under independent sampling of individuals, because the estimation windows described below ensure that no individual contributes to both $\widehat{\tau}_{2012}$ and $\widehat{\tau}_{2008\text{–}2011}$.

5.3 Estimation with a Discrete Running Variable

Month of birth takes only twelve distinct values in each estimation window, six on each side of the cutoff. With a running variable of this kind, the discontinuity at the cutoff must be extrapolated from a few support points. The bandwidth cannot shrink below the distance between support points, so the bias of a local polynomial estimator, its specification error, does not vanish as the sample grows, and inference that ignores it can be misleading however large the sample. A common response has been to cluster standard errors by values of the running variable (Lee and Card 2008). Kolesár and Rothe (2018) show that such confidence intervals do not guard against misspecification and can have poor coverage, and propose alternatives with guaranteed coverage under explicit restrictions on the conditional expectation function, closely related to the bias-aware construction of Armstrong and Kolesár (2020), in which the critical value is inflated to account for the largest bias consistent with a bound on the curvature of the conditional expectation function.

I estimate each discontinuity by local linear regression within a fixed window, using the RD-

Honest implementation of these methods using a triangular kernel, which assigns greater weight to observations closer to the September school-entry cutoff. It is important to be precise about which elements of this framework the paper uses. Because the window is fixed by the researcher rather than chosen by the procedure, the point estimates do not depend on the smoothness bound that the bias-aware procedure uses to select bandwidths and critical values: they are local linear regression estimates. The standard errors are the heteroskedasticity-robust standard errors produced by the procedure and are not clustered by month of birth, following the recommendation of Kolesár and Rothe (2018). Standard errors for the underlying regression discontinuity estimates are calculated using the Eicker–Huber–White heteroskedasticity-robust variance estimator, without clustering. The bias-aware critical values are not used for the main tests. The inference reported in this paper is therefore conventional inference applied to RDHonest point estimates and standard errors, not bias-aware inference, and it inherits the limitations set out in Section 5.4.

Two windows. The pooled estimator in (2) uses a 12-month window, six birth months on each side of each cutoff. With this window, the observations around cutoff t are the younger half of entry cohort $t-1$ and the older half of cohort t, so the windows around adjacent cutoffs are disjoint and no individual contributes to more than one discontinuity. A wider window would place the same individuals above one cutoff and below the next, so that the pooled pre-reform estimate would use some individuals twice and the two terms in (2) would share observations. The annual figures use a 24-month window, twelve months on each side, to display the evolution of the discontinuity across cohorts using the full cohorts on either side of each cutoff. Adjacent annual estimates in the figures therefore share an entire cohort and are not independent, and because they use a wider window than the pooled estimator, differences between plotted estimates need not reproduce the pooled estimates. The annual estimates are supporting visual and diagnostic evidence that simultaneously serve as a bandwidth check, since the formal test only has 12 mass points around the cutoff; formal estimates and tests come from (2).

5.4 Inference

The conventional test in Section 5.2 abstracts from two sources of error.

Smoothing bias. The test treats each discontinuity estimate as unbiased. With a discrete running variable this is an assumption, not an approximation that improves with sample size. Bias-aware confidence intervals of the kind proposed by Armstrong and Kolesár (2020) and Kolesár and Rothe (2018) avoid the assumption, but with six support points on each side of the cutoff the worst-case bias they must accommodate is large relative to sampling error, and bounding the bias of each discontinuity separately compounds the problem for a difference between two discontinuities. Such intervals can therefore be much wider than conventional ones. The difference-in-discontinuities design offers a reason to expect the relevant bias to be smaller than these worst-case bounds: the bias of each local linear estimate depends on the curvature of the conditional expectation function near the cutoff, and to the extent that the relative-age profile has a similar shape in the 2012 and pre-reform cohorts, the biases of the two terms in (2) are similar and difference out. This is a maintained assumption, not a result. The reported p-values are valid under it, and do not carry the coverage guarantees of bias-aware procedures.

Cohort-level shocks. The standard errors reflect sampling variation across individuals within each window, not shocks common to an entire cohort. Equation (3) shows that cohort-level shifts that do not follow a trend enter Δ directly. Because the 2011 cohort contributes to both terms of (2), a shock to that cohort moves them in opposite directions; the zero-covariance assumption is exact under independent sampling of individuals but not in the presence of such shocks. With a single treated cohort, the variance of cohort-level shocks cannot be estimated from the treated side, and cluster-robust and wild-bootstrap procedures are unreliable with one treated cluster (Cameron et al. 2008; MacKinnon and Webb 2018; Conley and Taber 2011). The annual estimates provide an informal check: if the pre-reform discontinuities vary across cohorts by more

than their standard errors imply, cohort-level shocks are material.

5.5 Contamination of the Final Pre-Reform Cohort

The problem. The fee a student paid depended on the year in which they entered university. Members of the 2011 entry cohort who delayed entry by a year, for example after a gap year, entered in 2012 under the new fees, as did members of the 2010 cohort who delayed by two years. Because the reform was announced in late 2010, members of the 2011 cohort could also respond in anticipation, either by entering on time rather than deferring or by abandoning plans for delayed entry. The nominally untreated 2011 cohort was therefore partially treated, and so the pre-reform counterfactual is contaminated.

Equation (3) shows why this matters. Suppose the higher fees deterred some of the 2011 cohort's would-be delayed entrants, lowering that cohort's cumulative entry by $\delta > 0$. This lowers μ_{2011} by δ and so raises Δ by $(1 + \omega_{2011})\delta$, biasing the estimated effect on cumulative entry toward zero. Most of this attenuation operates through the comparison side of the 2012 cutoff, because the individuals immediately below that cutoff belong to the 2011 cohort; the remainder operates through the pre-reform benchmark. Without a correction, the estimand for cumulative outcomes is therefore not the effect of the new fee regime relative to the old one. It is the effect of losing access to the old fee for on-time entry, relative to a group that retained it for on-time entry but faced the new fees if it delayed.

The correction is applied to the whole 2011 entry cohort, individuals born between September 1992 and August 1993. Within the 12-month windows, these individuals appear in two places: those born in the six months after the 2011 cutoff, above that cutoff in the pre-reform pool; and those born in the six months before the 2012 cutoff, on the comparison side of the treated discontinuity. Correcting the cohort as a whole therefore addresses both channels through which it enters equation (3). The correction takes two forms, depending on whether the outcome is defined for the whole cohort or only for university entrants.

Construction for unconditional outcomes. I estimate a logistic regression of each dependent variable on individual characteristics using the 2008–2010 cohorts. The covariates are month of birth, a female indicator, a LOPAR indicator, FSM eligibility, the number of GCSEs achieved at each grade (A*, A, B, C and so forth), school-LSOA fixed effects, and a linear cohort trend. Using the estimated model, I predict the outcomes that every member of the 2011 cohort would have had under the entry process of earlier cohorts, extending the cohort trend by one year, and use these predictions in place of the 2011 cohort's observed unconditional outcomes. The corrected estimate compares the 2012 cohort with a 2011 cohort whose outcomes reflect the earlier entry process. It targets θ, the effect of the new fee regime relative to the old one for individuals at the 2012 cutoff, under two assumptions. First, absent the reform, the 2011 cohort's outcomes given its observed characteristics would have followed the process estimated on earlier cohorts, with the trend extended by one year. This rules out departures from trend in the 2011 cohort that are unrelated to the reform, which the correction would wrongly remove, and relies on extrapolating a trend estimated over years that include the recession. Second, because the corrected outcomes enter discontinuity estimates in which the relative-age profile is the object of interest, the predictions must reproduce that profile. The model includes month of birth, so they reproduce the profile estimated on the 2008–2010 cohorts; the correction therefore assumes that the 2011 cohort's counterfactual profile matches theirs, the same stability assumption on which the design rests.

Reweighting for conditional outcomes. Outcomes such as the selectivity of the institution attended are defined only for university entrants. For these outcomes the problem is different: the reform's effect on the 2011 cohort changes who is observed entering, so the 2011 entrants may be selected differently from earlier entrants. I therefore reweight the members of the 2011 cohort

who entered university in 2011 by

$$w_i \;=\; \frac{\widehat{P}_{\text{pre}}(E_i = 1 \mid X_i)}{\widehat{P}_{2011}(E_i = 1 \mid X_i)}, \tag{5}$$

where E_i indicates on-time university entry and X_i are the covariates above. The numerator is the predicted probability of on-time entry from the model estimated on cohorts before 2011; the denominator is the corresponding prediction from a model with the same covariates estimated on the 2011 cohort (in which the cohort trend is not identified). To see what the weights do, let $f_{2011}(X)$ denote the distribution of characteristics in the 2011 cohort. Among 2011 entrants, characteristics are distributed in proportion to $P_{2011}(E = 1 \mid X) f_{2011}(X)$; had the same cohort been selected by the earlier process, they would have been distributed in proportion to $P_{\text{pre}}(E = 1 \mid X) f_{2011}(X)$. The weight is the ratio of the two, up to a constant, so the weighted 2011 entrants have the composition that the earlier selection process would have produced from the 2011 cohort, in the spirit of DiNardo et al. (1996). Entrants whose characteristics made them more likely to enter in 2011 than earlier entry patterns predict receive less weight; those whose entry became less likely receive more.

Reweighting addresses selection on observed characteristics. It recovers the counterfactual conditional outcome under the assumption that, given X_i, entrants' outcomes, such as where they enrolled and what they studied, are the same as they would have been under the earlier process. It does not address changes in where entrants with given characteristics enrolled, for instance students accepting less selective places to secure entry before the fee increase. Nor does it identify the full set of counterfactual outcomes for individuals who would have entered under the earlier process but did not: their characteristics are represented only through observed entrants with similar characteristics, which requires every type that would have entered under the earlier process to have a positive probability of entering in 2011.

Construction and reweighting are different operations. Unconditional outcomes are constructed; conditional outcomes are reweighted. Construction replaces the 2011 cohort's outcomes with what the earlier process predicts for the whole cohort. Reweighting leaves observed outcomes unchanged and alters only the relative weight of observed entrants. The two rest on different assumptions and target different objects, and the paper does not use them interchangeably.

Inference for corrected estimates. The standard errors for corrected estimates treat the fitted entry models as if their coefficients were known. Constructed outcomes are predicted probabilities, which vary less across individuals than realised binary outcomes, which is why the constructed estimates have smaller standard errors than the uncorrected ones; if the coefficients were known, this gain in precision would be legitimate. They are estimated, however, and the resulting prediction errors are shared: an error in the trend coefficient shifts the prediction for every member of the 2011 cohort, and an error in a school-LSOA fixed effect shifts it for every student in that school neighbourhood. Shared errors of this kind do not average out across the 2011 cohort, and the reported p-values do not reflect them. Nor do they reflect uncertainty about whether the 2011 cohort would have continued the trend of earlier cohorts, which is a modelling assumption rather than sampling error.[2]

Remaining contamination. Only the 2011 cohort is corrected, but every earlier cohort also contains individuals who delayed entry long enough to face the new fees: members of the 2010 cohort who delayed by two years and, at horizons of three to five years, members of earlier cohorts who delayed for longer. Exposure is therefore present in every pre-reform cohort and grows toward 2011. Its effect on the estimates depends on where each cohort sits in the design. Let δ_k denote the reduction in a cumulative outcome that this exposure causes in cohort k, and

[2]As an informal check, I evaluate each constructed estimate against the standard error of the corresponding uncorrected estimate, which is computed from realised outcomes. Every constructed estimate that is significant at the 5 percent level remains so. The check does not capture estimation error in the entry model.

suppose for the moment that it is the same for the cohort's older and younger members. Because cohort k supplies the treatment side of cutoff k and the comparison side of cutoff $k+1$, it lowers τ_k by δ_k and raises τ_{k+1} by δ_k, so each annual discontinuity changes by $\delta_{k-1} - \delta_k$, the difference in exposure between adjacent cohorts. With exposure rising toward 2011, the annual pre-reform discontinuities for cumulative outcomes should therefore drift downward In the pooled estimate, however, these differences should difference out. With equal weights, the pooled pre-reform discontinuity changes by $(\delta_{2007} - \delta_{2011})/4$, and once the 2011 cohort is corrected, by $\delta_{2007}/4$ alone: the exposure of the 2008–2010 cohorts cancels, because each of those cohorts appears once on each side of a pre-reform cutoff.[3] What remains is (i) the 2007 term, from members of the 2007 cohort who entered five years late, which is small and pushes the estimate away from zero; (ii) differences in exposure between a cohort's older and younger members, which attenuate the estimate if exposure is concentrated among older members, who sit above the earlier cutoff, and work in the opposite direction if it is concentrated among younger members; and (iii) departures from equal weighting in the pooled fit, which are small because the cohorts are of similar size. Remaining contamination is therefore a plausible but uncertain source of attenuation, and likely to be second-order relative to the contamination of the 2011 cohort, which enters with coefficient $-(1 + \omega_{2011})$ and is unambiguously attenuating. At the one-year horizon only the 2011 cohort could have entered under the new fees, so the corrected one-year estimate is not affected by remaining contamination from earlier cohorts. I do not interpret any corrected estimate as a bound.

[3]For example, if exposure lowered five-year entry by 0.1, 0.2 and 0.4 percentage points in the 2008, 2009 and 2010 cohorts and by 0.05 points in the 2007 cohort (illustrative values), then with the 2011 cohort corrected the discontinuities at the 2008–2011 cutoffs would change by -0.05, -0.1, -0.2 and $+0.4$ points. These sum to 0.05, so the pooled pre-reform discontinuity would change by about 0.01 points.

5.6 Threats to Identification and Covariate Balance

The identifying assumption is that, apart from the reform, nothing changed the discontinuity at the 2012 cutoff relative to the pre-reform discontinuity. Equation (3) makes clear that the threat is a shock specific to the 2011 or 2012 cohort that departs from the pre-reform trend: concurrent policies, changes in the composition of births around the cutoff, or changes in the timing of school entry.

Covariate balance. Appendix Table A.1 reports regression discontinuity estimates for three predetermined characteristics at each cutoff from 2008 to 2012. The results are mixed. There is no statistically significant discontinuity in the share female in any year. For FSM eligibility, the reported confidence intervals exclude zero in 2008 (-2.9 percentage points) and 2010 (-1.1), and for residence in a low-participation area they exclude zero in 2011 (-1.5). None of the three discontinuities is significant at the 2012 cutoff. Balance therefore does not hold in every pre-reform cohort. One possible reason is that characteristics such as FSM eligibility are recorded at a fixed school stage, so the two sides of each cutoff are measured a year apart, and changes in eligibility or recording between consecutive years appear as discontinuities.

Individually significant pre-reform discontinuities do not by themselves invalidate the design, which requires that the discontinuity at the 2012 cutoff equal the pooled pre-reform discontinuity in the absence of the reform, not that every annual discontinuity be zero. What matters is whether the covariate discontinuities differ between 2012 and the pre-reform period. Relative to the average of the four annual pre-reform estimates, the 2012 discontinuity is about 1.2 percentage points higher for FSM eligibility, 1.0 points higher for low-participation residence, and 0.5 points lower for the share female. To gauge whether differences of this size matter, I multiply each by the raw difference in the outcome between the corresponding groups in the 2012 estimation window. For five-year entry, the implied compositional contributions sum to about -0.3 percentage points, and for on-time entry to about -0.2 points, in both cases in the direction of a larger estimated

contraction; for unconditional top-quartile attendance and STEM participation they are about -0.1 points. Because FSM eligibility and low-participation residence are correlated, summing their contributions overstates the combined effect, and the calculation uses rounded estimates; it indicates only that compositional differences of the observed size could account for a modest part of the enrolment estimates and are unlikely to account for their sign or most of their magnitude

Strategic delay of school entry. Some summer-born children defer school entry to avoid the relative age effect, a practice concentrated among higher-income families. For this to bias the results, there would need to be a cohort-specific change in delayed school entry for children who would later face the 2012 fees, that is, among children starting school around 1998. Appendix Figure A.6 plots the share of students delaying primary school entry by university-entry cohort: the proportion is stable at approximately 0.5 percent, with no discontinuity around the treated cohort. The main analysis excludes individuals whose school-entry timing does not align with their birthdate cohort

Seasonality of birth. Family background varies with season of birth (Buckles and Hungerman 2013), and the relative age premium partly reflects this selection. The design does not require births to be random across months, only that the selection pattern be stable across cohorts, in which case it is differenced out; the balance evidence above bears on this.

Concurrent policies. The raising of the participation age affects later cohorts, and changes to maintenance support were incremental (Section 3.3). The AAB exemption from number controls applied to exactly the cohort treated by the fee increase and is examined directly in Section 6.4.

6 Results

All estimates are pooled difference-in-discontinuities estimates from (2), expressed in percentage points. Significance refers to the conventional test described in Section 5.2, subject to the caveats in Section 5.4.

6.1 University Enrolment: Immediate Contraction, Delay, and Persistence

Table 2 reports estimates for on-time entry and for cumulative entry within one to five years of the on-time date, without (column 1) and with (column 2) the correction for the 2011 cohort. Figure 1 plots the annual discontinuities for on-time entry and entry within five years, and Appendix Figure A.7 those for the intermediate horizons, which show a similar pattern.

On-time entry. The corrected estimate implies that the reform reduced on-time university entry by 4.57 percentage points; the uncorrected estimate is 4.02 points. Both are highly significant. For scale, 24.8 percent of the 2011 cohort entered university on time (Table 1), so the corrected estimate corresponds to roughly 18 percent of the on-time entry rate.

The correction increases the magnitude of the on-time estimate. Anticipatory entry by the 2011 cohort, entering on time rather than deferring to avoid the higher fees, would inflate that cohort's on-time entry and hence the magnitude of the uncorrected estimate, so a correction for anticipation alone would reduce it. That the correction instead increases the estimate indicates that, relative to the entry process of earlier cohorts, the 2011 cohort's on-time entry was, if anything, lower than predicted. This is consistent with the entry controls in force from 2009–10 (Higher Education Funding Council for England 2011) limiting the extent to which additional demand could translate into additional entry; it may also reflect the extrapolation of the cohort trend.

Table 2: The Effect of the Reform on University Entry

	(1) Uncorrected	(2) 2011 cohort constructed
On-time entry	-4.02^{***}	-4.57^{***}
	[<0.001]	[<0.001]
Entered within 1 year	-1.14^{**}	-2.51^{***}
	[0.040]	[<0.001]
Entered within 2 years	-1.15^{**}	-2.15^{***}
	[0.043]	[<0.001]
Entered within 3 years	-1.16^{**}	-2.11^{***}
	[0.045]	[<0.001]
Entered within 4 years	-1.08^{*}	-2.22^{***}
	[0.062]	[<0.001]
Entered within 5 years	-1.05^{*}	-2.21^{***}
	[0.072]	[<0.001]
N, pre-reform cutoffs (2008–2011)	986,275	986,275
N, 2012 cutoff	242,865	242,865

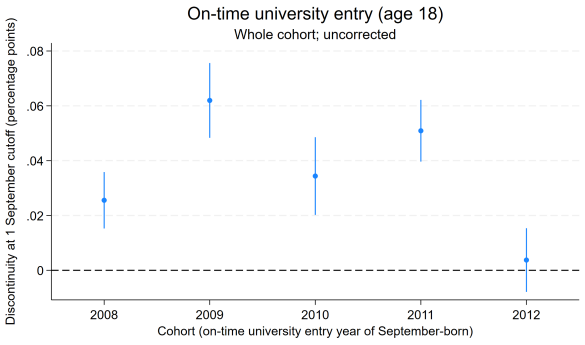
Notes: Each cell reports $\widehat{\Delta} = \widehat{\tau}_{2012} - \widehat{\tau}_{2008\text{–}2011}$ in percentage points (equation 2), where each discontinuity is a local linear RDHonest estimate within a 12-month window (six birth months on each side of the 1 September cutoff) and the pre-reform discontinuity is estimated on the pooled observations from the 2008–2011 cutoffs. Outcomes are defined for the whole cohort. Column (2) replaces the outcomes of the 2011 entry cohort with outcomes constructed from the entry process of earlier cohorts (Section 5.5). P-values in brackets are from a conventional two-sided test using $SE(\widehat{\Delta}) = [SE(\widehat{\tau}_{2012})^2 + SE(\widehat{\tau}_{2008\text{–}2011})^2]^{1/2}$; they are not bias-aware, and those in column (2) do not account for estimation of the entry model. The pre-reform sample is 986,270 for on-time and five-year entry. Counts rounded to the nearest five. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

Delayed and persistent effects. The corrected estimates for cumulative entry are -2.51 percentage points within one year of the on-time date, and between -2.11 and -2.22 points at horizons of two to five years; all are significant at the 0.1 percent level. About 45 percent of the immediate contraction was therefore reversed by entry one year late, little further catch-up occurred over the following four years, and at five years the gap stood at 2.21 points, 48 percent of the on-time effect. Postponement and the absence of entry by age 23 each account for roughly half of the immediate response.

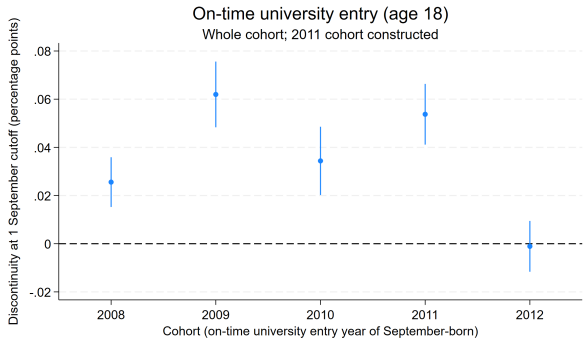
The uncorrected estimates are smaller. Within one to three years they range from -1.14 to -1.16 points (p between 0.040 and 0.045); at four and five years they are -1.08 ($p = 0.062$) and -1.05 ($p = 0.072$), not significant at the 5 percent level. Read at face value, they would suggest that about three-quarters of the on-time response was postponement and that the remainder cannot be distinguished from zero at five years. The difference between the two sets of estimates opens at the one-year horizon and remains around one percentage point thereafter. This is the pattern expected if the uncorrected estimates are attenuated by the 2011 cohort's treated delayed entry, which first affects cumulative outcomes at the one-year horizon. Because only the 2011 cohort could have entered under the new fees within one year, the corrected one-year estimate is the cumulative estimate least exposed to remaining contamination.

Postponement versus non-entry. The distinction between the on-time and five-year estimates is between postponing entry and not having entered by age 23. The evidence indicates that much of the immediate response was postponement, but also that the reform reduced the share of the cohort entering university by 23. It does not establish that this reduction is permanent: some of those who had not entered by 23 may enter later, for example as mature students, and the horizon is limited by the data. What the estimates do show is that the reduction was not undone within five years of the on-time date, and that the corrected estimates show no further catch-up after the first year. The two responses have different interpretations. Postponement is consistent with

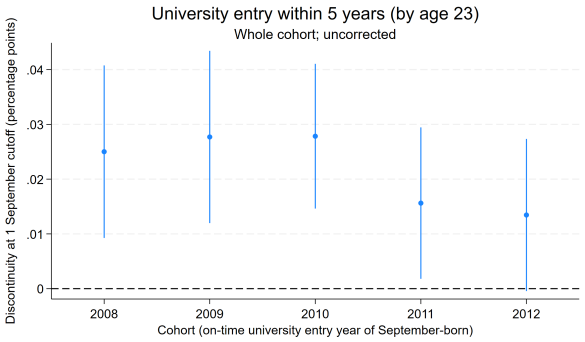
Figure 1: University Enrolment: Annual Discontinuities, 2008–2012



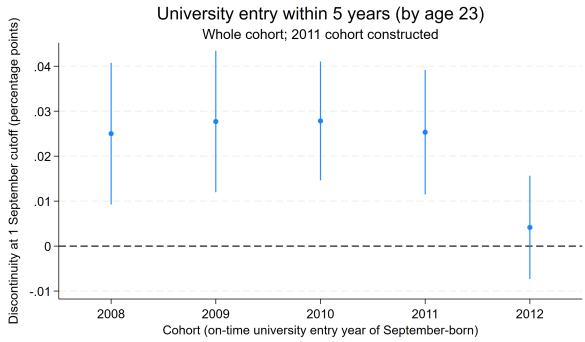
(a) On-time entry, original



(b) On-time entry, 2011 cohort constructed



(c) Entry within five years, original



(d) Entry within five years, 2011 cohort constructed

Note: Outcomes are defined for the whole cohort: on-time entry is entry in the academic year following the individual's 18th birthday, and entry within five years is entry by age 23. Panels (a) and (c) use observed outcomes; panels (b) and (d) use outcomes for the 2011 entry cohort constructed from the entry process of earlier cohorts (Section 5.5). Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

students re-optimising the timing of entry, working or taking time out before committing to a more expensive degree. The absence of entry by 23 is consistent with the fee increase moving some marginal students out of higher education over the horizon observed.

How conservative is the corrected estimate? Any treated delayed entry in the 2011 cohort that the correction fails to remove biases the estimates toward zero, and contamination of earlier cohorts can attenuate them further if it is concentrated among their older members (Section 5.5). These considerations make it plausible that the corrected five-year estimate understates the persistent contraction. But the correction itself rests on extrapolating the entry process of earlier cohorts, and if that extrapolation overstates the entry the 2011 cohort would have achieved without the reform, the corrected estimate overstates the contraction. I therefore treat the corrected estimate as the preferred estimate rather than as a bound, and read the two columns of Table 2 together: the uncorrected estimates are biased toward zero by contamination that the design shows must attenuate them, while the corrected estimates depend on the counterfactual model.

6.2 Where Entrants Enrolled: Conditional Selectivity

Panel A of Table 3 and Figure 2 report effects on the selectivity of the institution attended, conditional on university entry. Without reweighting, entrants became 4.41 percentage points more likely to attend an above-mean institution and 3.74 points more likely to attend a top-quartile institution (both $p < 0.001$). After reweighting the 2011 entrants, the estimates are 5.99 and 5.27 points. The estimates at the top 20 percent (2.50 unweighted, 3.87 reweighted) and top 10 percent (1.47 and 2.06) are smaller but also significant at the 1 percent level. Relative to the shares of entrants attending such institutions in the 2012 estimation window, about 39 percent above the median and 21 percent in the top quartile, the reweighted estimates correspond to increases of roughly 15 and 25 percent.

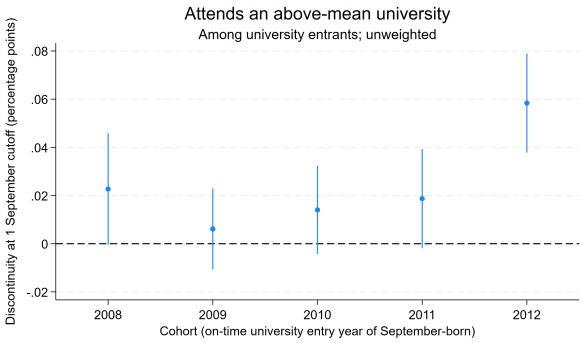
Reweighting gives less weight to 2011 entrants whose entry became more likely than earlier

Table 3: The Effect of the Reform on the Selectivity of the Institution Attended

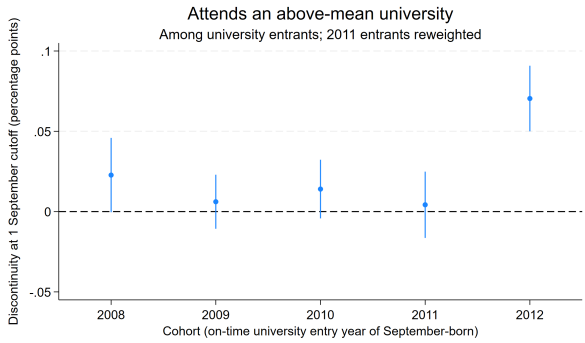
Panel A. Conditional on university entry		
	(1)	(2)
	Unweighted	2011 entrants reweighted
Above-mean institution	4.41^{***}	5.99^{***}
	[<0.001]	[<0.001]
Top-quartile institution	3.74^{***}	5.27^{***}
	[<0.001]	[<0.001]
Top 20% institution	2.50^{***}	3.87^{***}
	[0.002]	[<0.001]
Top 10% institution	1.47^{***}	2.06^{***}
	[0.003]	[<0.001]
N, pre-reform / 2012	345,055 / 87,005	345,055 / 87,005
Panel B. Whole cohort (unconditional)		
	(1)	(2)
	Uncorrected	2011 cohort constructed
Above-mean institution	1.23^{***}	0.52^{*}
	[0.004]	[0.097]
Top-quartile institution	1.14^{***}	0.86^{***}
	[<0.001]	[<0.001]
Top 20% institution	0.75^{**}	0.71^{***}
	[0.011]	[0.001]
Top 10% institution	0.48^{***}	0.42^{***}
	[0.007]	[<0.001]
N, pre-reform / 2012	986,275 / 242,865	986,275 / 242,865

Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table 2. Institutions are ranked by the mean A-level points of their entrants in a pre-reform reference year. Panel A outcomes are defined among university entrants; column (2) reweights the 2011 entrants by equation (5). Panel B outcomes are defined for the whole cohort and equal zero for individuals who do not enter; column (2) replaces the outcomes of the 2011 cohort with constructed outcomes. P-values in brackets are from the conventional two-sided test; they are not bias-aware and do not account for estimation of the entry models. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

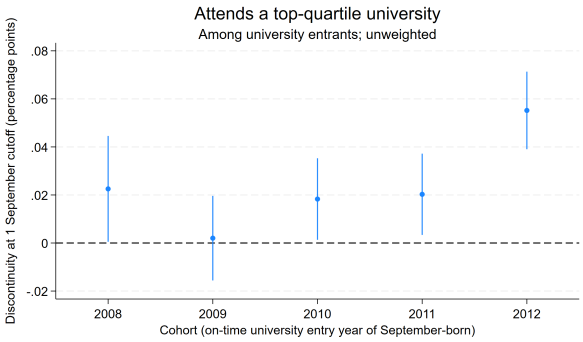
Figure 2: Selectivity of the Institution Attended, Conditional on Entry: Annual Discontinuities



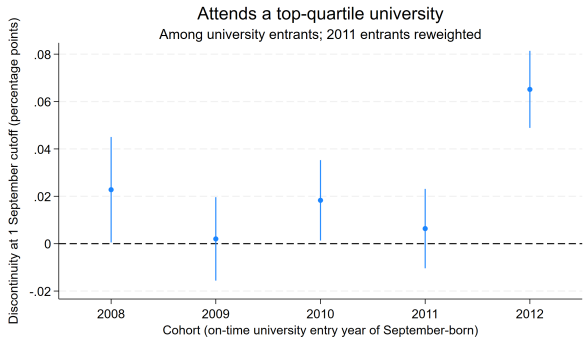
(a) Above-mean institution, unweighted



(b) Above-mean institution, reweighted



(c) Top-quartile institution, unweighted



(d) Top-quartile institution, reweighted

Note: Outcomes are indicators for attending an institution above the median, or in the top quartile, of the selectivity distribution, defined among individuals who enter university. Panels (a) and (c) are unweighted; panels (b) and (d) reweight the 2011 entrants to the composition implied by the entry process of earlier cohorts (equation 5). Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

patterns predict, and more to those whose entry became less likely. That the reweighted estimates are larger therefore indicates that, relative to the earlier selection process, the 2011 entrant pool was tilted toward students who would attend more selective institutions, which raises the comparison cohort's selectivity and attenuates the unweighted estimates. This is consistent with entry in 2011 having been rationed toward stronger applicants, or with anticipatory responses concentrated among students bound for selective institutions; the reweighting does not distinguish these explanations.

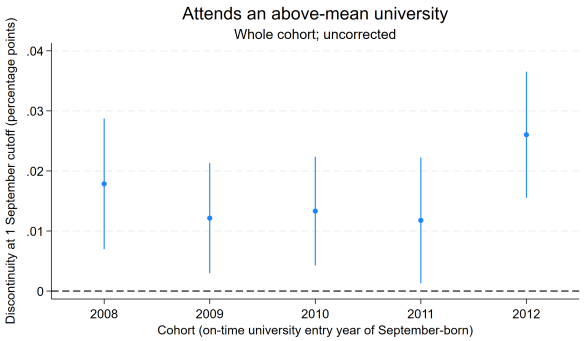
Appendix Figure A.5 shows the raw share of entrants attending selective institutions over time. Descriptively, we see a sharp jump in the percentage of individuals selecting into higher ranked institutions after 2012. The conditional estimates describe a substantial change in the composition of university attendance among entrants. They do not describe a change in the experience of the cohort as a whole, because the population of entrants itself changed. Section 6.3 turns to the cohort.

6.3 Selectivity Across the Whole Cohort

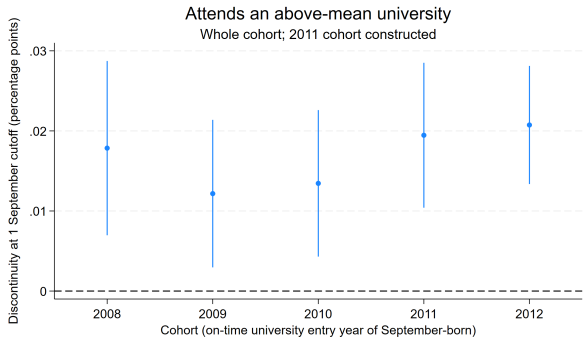
Panel B of Table 3 and Figure 3 report effects on the share of the whole cohort attending institutions of each level of selectivity. The uncorrected estimates are 1.23 percentage points for above-median institutions ($p = 0.004$), 1.14 for the top quartile ($p < 0.001$), 0.75 for the top 20 percent ($p = 0.011$), and 0.48 for the top 10 percent ($p = 0.007$). The correction reduces the above-median estimate by more than half, to 0.52 points ($p = 0.097$), which is not significant at the 5 percent level. The corrected estimates at the top quartile (0.86, $p < 0.001$), top 20 percent (0.71, $p = 0.001$), and top 10 percent (0.42, $p < 0.001$) remain positive and significant. There is therefore evidence of an increase in attendance at the most selective institutions across the cohort, but it is much smaller than the conditional estimates.

The conditional and unconditional estimates are linked mechanically. The conditional share is the unconditional share divided by the entry rate, so a fall in entry raises the conditional share

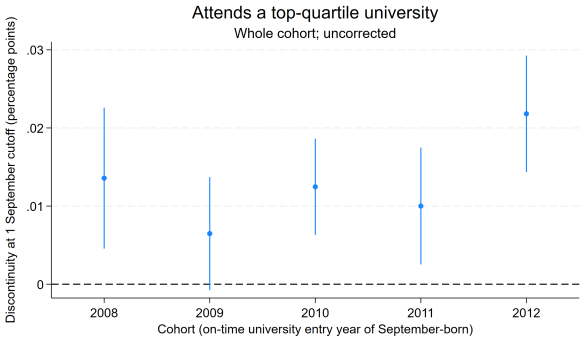
Figure 3: Selectivity of the Institution Attended, Whole Cohort: Annual Discontinuities



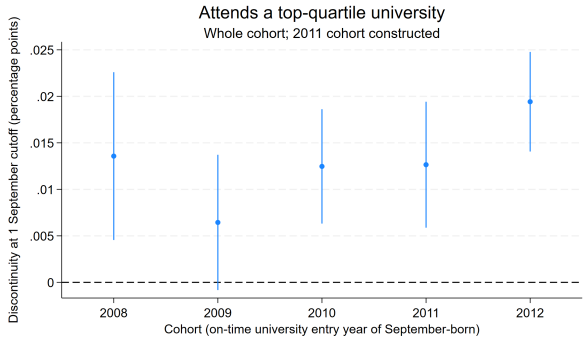
(a) Above-mean institution, original



(b) Above-mean institution, 2011 cohort constructed



(c) Top-quartile institution, original



(d) Top-quartile institution, 2011 cohort constructed

Note: Outcomes are indicators for attending an institution above the median, or in the top quartile, of the selectivity distribution, defined for the whole cohort and equal to zero for individuals who do not enter university. Panels (a) and (c) use observed outcomes; panels (b) and (d) use outcomes for the 2011 entry cohort constructed from the entry process of earlier cohorts (Section 5.5). Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

even if no one changes institution. In the 2012 estimation window, about 36 percent of the cohort entered university within five years, and about 39 percent of entrants attended an above-median institution. If the 2.2-point fall in participation had fallen entirely on institutions below the mean, with no change in the number of the cohort attending above-median institutions, the share of entrants attending above-median institutions would have risen by about 2.6 points. Adding the corrected unconditional increase of 0.5 points raises this to about 4 points. Selection out of higher education alone can therefore account for a large part of the conditional estimates of 4.4 (unweighted) and 6.0 (reweighted) points.

The unconditional estimates also locate the contraction. Five-year entry and unconditional above-mean attendance are estimated on the same individuals with an estimator that is linear in the outcome, so the difference between the two estimates is the estimated effect on entering a below-mean institution. The corrected point estimates imply a fall of about 2.7 percentage points in entry to below-median institutions and a fall of about 0.3 points in entry to institutions between the median and the top quartile, alongside the increase of 0.86 points at top-quartile institutions. The reform thus reduced the number of entrants while modestly increasing the number of the cohort attending the most selective quarter of institutions. The fall in entry was concentrated among students who would have attended less selective institutions; those who continued were a group more likely to attend selective institutions in any case; and a small number more of the cohort attended highly selective institutions than would otherwise have done so. The conditional estimates combine all three.

6.4 The AAB Exemption

The concurrent exemption of AAB+ students from student number controls freed selective institutions to recruit more high-attaining students in 2012–13, and could have contributed to the shift toward selective institutions independently of the fee increase. Table 4 and Figure 4 repeat the analysis excluding students who achieved AAB or higher. These estimates are uncorrected

and unweighted, so column (1) of Table 4 reports the corresponding uncorrected full-sample estimates for comparison.

Table 4: Excluding Students Covered by the AAB Exemption

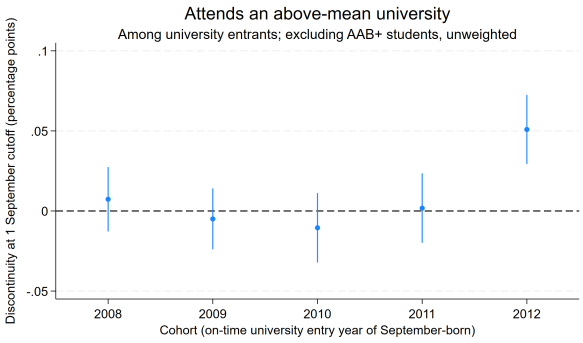
	(1) Full sample	(2) Excluding AAB+
On-time entry	-4.02^{***}	-3.33^{***}
	[<0.001]	[<0.001]
Conditional on university entry		
Above-median institution	4.41^{***}	5.22^{***}
	[<0.001]	[<0.001]
Top-quartile institution	3.74^{***}	4.41^{***}
	[<0.001]	[<0.001]
Top 20% institution	2.50^{***}	3.12^{***}
	[0.002]	[<0.001]
Top 10% institution	1.47^{***}	0.74^{**}
	[0.003]	[0.012]
Whole cohort (unconditional)		
Above-median institution	1.23^{***}	[pending]
	[0.004]	
Top-quartile institution	1.14^{***}	[pending]
	[<0.001]	

Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table 2. All estimates are uncorrected and unweighted; the AAB+ exclusion has not yet been combined with the construction or reweighting of the 2011 cohort. Column (2) excludes students achieving AAB or higher at A-level. Sample sizes (pre-reform / 2012): on-time entry 986,270 / 242,865 in column (1) and 927,160 / 227,260 in column (2); conditional outcomes 345,055 / 87,005 and 299,750 / 75,265; unconditional outcomes 986,275 / 242,865 in column (1). P-values in brackets are from the conventional two-sided test and are not bias-aware. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

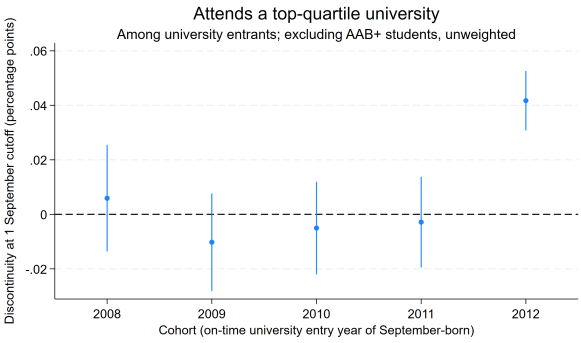
Excluding AAB+ students, on-time entry fell by 3.33 percentage points. Conditional on entry, the probability of attending an above-mean institution rose by 5.22 points, a top-quartile institution by 4.41 points, and a top-20-percent institution by 3.12 points (all $p < 0.001$); the top-10-percent estimate is 0.74 points ($p = 0.012$). The first three estimates are at least as large as the corresponding uncorrected full-sample estimates (4.41, 3.74 and 2.50). The selectivity response is therefore not confined to students who benefited from the exemption, which is important evidence against the exemption being the sole explanation for the rise in conditional selectivity. The

top-decile estimate is about half the full-sample estimate (1.47), consistent with the exemption playing a larger role at the most selective institutions, where, by construction of the ranking, high-attaining entrants are concentrated. Panels (c) and (d) of Figure 4 show the annual discontinuities for the unconditional outcomes; the pooled estimates are pending.

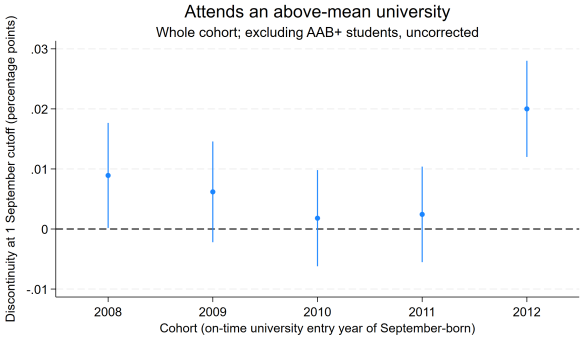
Figure 4: Excluding Students Covered by the AAB Exemption: Annual Discontinuities



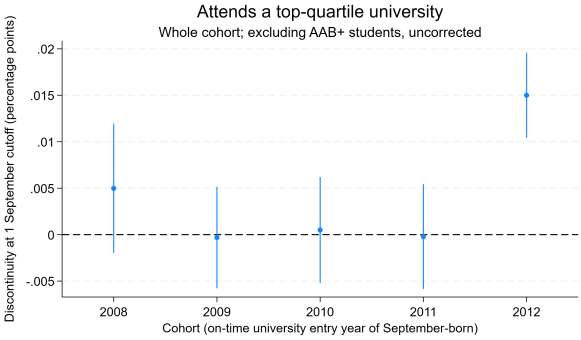
(a) Above-mean institution, conditional



(b) Top-quartile institution, conditional



(c) Above-mean institution, unconditional



(d) Top-quartile institution, unconditional

Note: The sample excludes students achieving AAB or higher at A-level. Panels (a) and (b) are defined among university entrants; panels (c) and (d) are defined for the whole cohort and equal zero for individuals who do not enter. All panels use observed, unweighted outcomes; the construction and reweighting of the 2011 cohort have not been applied. Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

Excluding AAB+ students does not isolate the fee reform perfectly. First, the policy applied to students with AAB "or equivalent" qualifications (Higher Education Funding Council for England 2011), whereas the exclusion here is based on A-level grades, so some exempt students may remain in the sample. Second, A-level results were obtained after the reform was announced; if

the reform affected attainment around the AAB threshold, excluding AAB+ students selects the sample on a potentially affected outcome. Third, the exemption and the accompanying reallocation of places could affect non-exempt students indirectly. Unrestricted recruitment of exempt students may have changed competition for controlled places, while the margin of up to 20,000 places was reserved for institutions charging average net fees of £7,500 or less (Higher Education Funding Council for England 2011), which, if anything, moved controlled places away from institutions charging the maximum fee. Equilibrium or institutional responses of this kind remain possible explanations for part of the response.

6.5 Subject Choice, Completion, and Progression

Table 5 reports estimates for STEM participation, degree completion, withdrawal for poor academic performance, and postgraduate study; the annual discontinuities are in Appendix Figures A.10 and A.11.

STEM. Across the whole cohort, STEM participation fell by 1.53 percentage points after correction ($p < 0.001$); the uncorrected estimate is -0.70 ($p = 0.081$). Among entrants, the estimates are -0.96 unweighted and -0.77 reweighted, neither statistically significant. The STEM share among entrants is about one-third in the 2012 estimation window, so the 2.2-point fall in participation alone would reduce unconditional STEM participation by about 0.75 points, roughly half of the corrected estimate. The negative but imprecise conditional estimates could account for part of the remainder. The decline in STEM participation across the cohort is therefore consistent with lower participation, and possibly a change in the composition of entrants, rather than with a substantial change in the probability of choosing STEM among those who entered. The conditional estimates are imprecise, however, and cannot rule out moderate changes in subject choice.

Table 5: Subject Choice, Completion, and Progression

	(1) Uncorrected / unweighted	(2) Corrected / reweighted	(3) N pre-reform	(4) N 2012
Panel A. Whole cohort (column 2: 2011 cohort constructed)				
STEM subject	-0.70^{*}	-1.53^{***}	986,275	242,865
	[0.081]	[<0.001]		
Panel B. Conditional on university entry (column 2: 2011 entrants reweighted)				
STEM subject	-0.96	-0.77	349,390	88,035
	[0.326]	[0.428]		
Postgraduate study	0.51	0.68	349,390	88,035
	[0.328]	[0.186]		
Degree completion	-1.26	-0.91	347,160	87,200
	[0.133]	[0.285]		
Withdrawal for poor academic performance	-1.27^{*}	-0.90	316,885	79,580
	[0.069]	[0.204]		

Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table 2. Withdrawal for poor academic performance is constructed from the HESA reason for ending a course and is not a general measure of non-completion. P-values in brackets are from the conventional two-sided test; they are not bias-aware and do not account for estimation of the entry models. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Completion, withdrawal, and postgraduate study. The point estimates for degree completion are -1.26 (unweighted) and -0.91 (reweighted) percentage points; for withdrawal for poor academic performance, -1.27 and -0.90; and for postgraduate study, 0.51 and 0.68. None is significant at the 5 percent level (the unweighted withdrawal estimate has $p = 0.069$). These estimates do not establish a statistically significant change in completion, in this measure of withdrawal, or in postgraduate study, which is not the same as establishing that there was no change: they are imprecise and cannot rule out economically meaningful effects. In particular, the evidence does not indicate that the reform increased withdrawal for poor academic performance. Two features of the data limit these outcomes. Delayed entrants have less time to complete a degree or to progress to postgraduate study before the end of the data, and the treated cohort delayed entry more; and postgraduate study is affected by the introduction of postgraduate loans in 2016.

7 Discussion: Participation, Timing, Selection, and Sorting

The results separate five margins of response that aggregate statistics combine. *University entry decisions*: on-time entry fell by 4.6 percentage points. *Delayed entry*: about half of that response was reversed by later entry, almost all of it within a year. *Selection into the attending sample*: participation within five years was 2.2 points lower, so entrants were a smaller and differently selected group. *Sorting between institutions*: among entrants, attendance at above-median and top-quartile institutions rose by roughly 4 to 6 points, but across the cohort the increase was below one point and concentrated in the top quartile, while entry to institutions below the median fell. *Subject choice among entrants*: the STEM share among entrants did not change detectably, while STEM participation across the cohort fell, in line with lower participation. Changes in the distribution of outcomes among entrants cannot be read as changes for the underlying population: an analysis of entrants alone would find a pronounced upgrading of institutional quality and no

change in subject choice, while the cohort as a whole experienced a fall in participation, a modest increase in attendance at the most selective institutions, and a fall in STEM participation.

Mechanisms for the selectivity response. Several mechanisms are consistent with the evidence, and the design does not distinguish between them. The first is selection: a common increase in price deters the entrants for whom university is least attractive, and if these are disproportionately students who would have attended less selective institutions, the conditional share at selective institutions rises even if no individual changes destination. Section 6.3 shows that this mechanism alone can account for a large part of the conditional estimates. The second is greater attention to expected returns: a higher price may lead students who do enter to weigh more carefully where they enrol, raising relative demand for institutions perceived to offer higher returns. The increase in unconditional attendance at top-quartile institutions requires some mechanism beyond selection, because selection out of higher education cannot raise the share of the cohort attending a given institution. The third is supply and admissions. Number controls, the AAB exemption, and the reallocation of controlled places all changed in 2012, and institutions may have responded to the new fee regime in their recruitment. A reduction in delayed entry from the previous cohort could also have freed places at the institutions those students would have attended. The persistence of the response among students below the AAB threshold indicates that the exemption is not the sole explanation, but does not rule out other supply-side responses.

One feature of the English system makes the returns-based interpretation less straightforward than it may appear. Under income-contingent repayment with a write-off after 30 years, the expected cost of a higher fee rises with expected graduate earnings, because graduates with low earnings repay only part of their balance (Britton et al. 2017). For a student who responds only to expected repayments, the fee increase therefore raised the effective price of options with higher expected earnings by more than that of options with lower expected earnings. A shift toward selective institutions is more naturally explained by responses to the headline price or the size

of the debt (Callender and Jackson 2005), by selection, or by changes on the supply side, than by a model in which students respond only to expected repayments. The results are consistent with students becoming more selective or returns-conscious, but they do not measure students' preferences or expectations directly.

Policy relevance. Income-contingent loans removed upfront costs, but they did not neutralise the participation response: the reform produced a large immediate contraction and a smaller but persistent one over five years among the individuals studied. The results also caution against judging such reforms by the characteristics of entrants. A rise in the share of entrants attending selective institutions is compatible with a fall in participation that is concentrated among students who would have attended less selective institutions, and the cohort-level reallocation was much smaller than the change among entrants. The results do not establish welfare improvements or identify individual net returns to higher education; whether the reallocation and the reduction in participation raised or lowered welfare depends on the returns to the education forgone and to the institutions attended.

8 Robustness and Limitations

Inference. The main tests are conventional tests applied to RDHonest estimates. They are not bias-aware, they do not capture cohort-level shocks, and for corrected estimates they do not account for estimation of the entry models (Sections 5.4 and 5.5). The on-time estimates are large relative to their standard errors under either version of the outcome; the longer-horizon and unconditional selectivity estimates are smaller and more dependent on the maintained assumptions.

Correction for the 2011 cohort. The corrected estimates depend on a model of the entry process estimated on earlier cohorts and extrapolated by one year. The paper reports corrected and uncorrected estimates side by side for this reason.

External validity. The estimates are effects at the boundary between two cohorts, for individuals born around 1 September, of a single large reform in one system. Every member of the 2012 cohort faced the new fees, and there is no particular reason to expect individuals at the boundary to respond differently from the rest of the cohort, but relative age is correlated with attainment, and the response may vary with it.

9 Conclusion

This paper asks what happened to university participation, and to where students enrolled, when the price of university in England nearly tripled under a system of universal income-contingent loans. Exploiting the alignment of the 2012 fee reform with the September school-entry cutoff, and comparing the discontinuity at the first treated cutoff with the pooled discontinuity at the four preceding cutoffs, I estimate the reform's effects on entry and on the selectivity of the institutions attended. Because fees depended on the year of entry, the final pre-reform cohort was partially exposed to the reform, and I correct its outcomes using the entry process of earlier cohorts.

Three findings emerge. First, the reform produced a significant contraction in university participation. On-time entry fell by 4.6 percentage points. About half of this response was reversed by delayed entry, almost entirely within a year, but participation within five years remained 2.2 points lower. The reform therefore postponed entry for some students and, over the horizon observed, prevented it for others. Without the correction for the contaminated comparison cohort, the persistent effect is about half as large and imprecise.

Second, the selectivity of entrants changed substantially. Conditional on entering university, students were 4.4 and 3.7 percentage points more likely to attend above-median and top-quartile institutions, and 6.0 and 5.3 points more likely after reweighting the final pre-reform cohort's entrants. The response is present among students not covered by the concurrent exemption of high attainers from number controls. It is consistent with a response to the higher cost of univer-

sity, potentially involving greater attention to expected returns, but also with selection and with changes on the supply side.

Third, the selectivity of entrants is a poor guide to the experience of the cohort. Across the whole cohort, attendance at top-quartile institutions rose by less than one percentage point, the effect on above-median attendance is small and imprecise, and the point estimates imply that entry to below-median institutions fell. STEM participation across the cohort fell, while subject choice among entrants did not change detectably. Separating entry, delay, selection into university, sorting between institutions, and subject choice is necessary to understand the reform, because aggregate changes cannot be inferred from the characteristics of students who continue into higher education.

The estimates rest on conventional inference applied to discontinuity estimates with a discrete running variable, on a model-based correction for the final pre-reform cohort, and on the assumption that the pre-reform discontinuity is the right counterfactual for 2012. Within those limits, the evidence indicates that a large increase in the price of university, even when deferred through income-contingent loans, reduced and delayed participation and changed who entered higher education and where.

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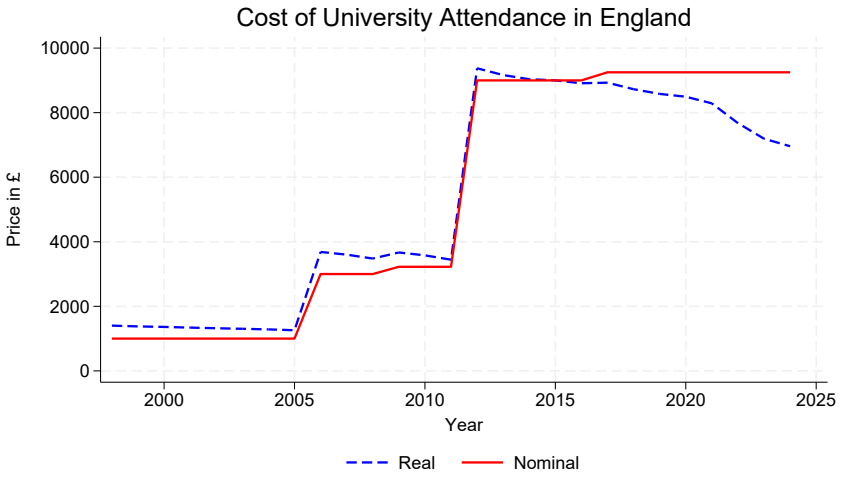
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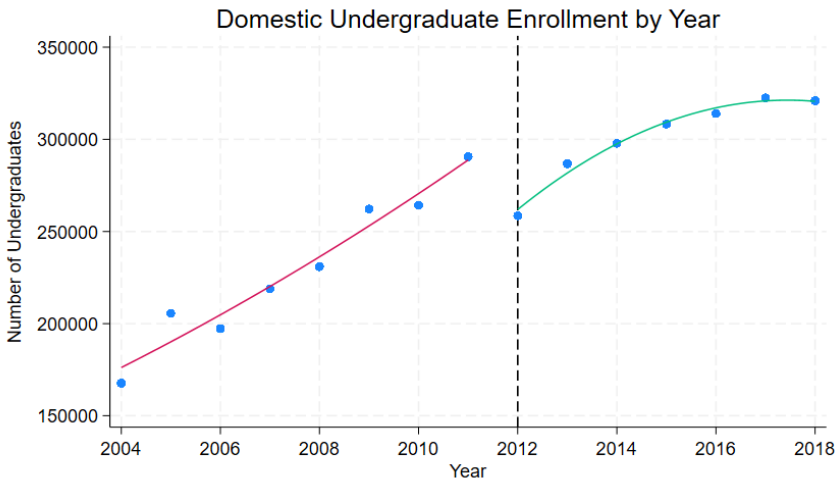
Appendix Figures and Tables

Figure A.1: Maximum Annual Tuition Fee, England



Note: Figure shows the maximum annual tuition fee chargeable to home undergraduate students in England, 1998–2024, in nominal and real (2024) prices. The 1998, 2006, and 2012 discontinuities correspond to the Teaching and Higher Education Act (1998), the Higher Education Act (2004), and the post-Browne reform respectively.

Figure A.2: Domestic First-Year Undergraduate Entrants by Year of Entry



Note: Figure shows the number of domestic first-year undergraduate entrants at English institutions by academic year of entry, 2004–2018. The dashed line marks the 2012–13 academic year, the first cohort subject to the £9,000 fee cap. Fitted trends are shown separately for the pre- and post-reform periods.

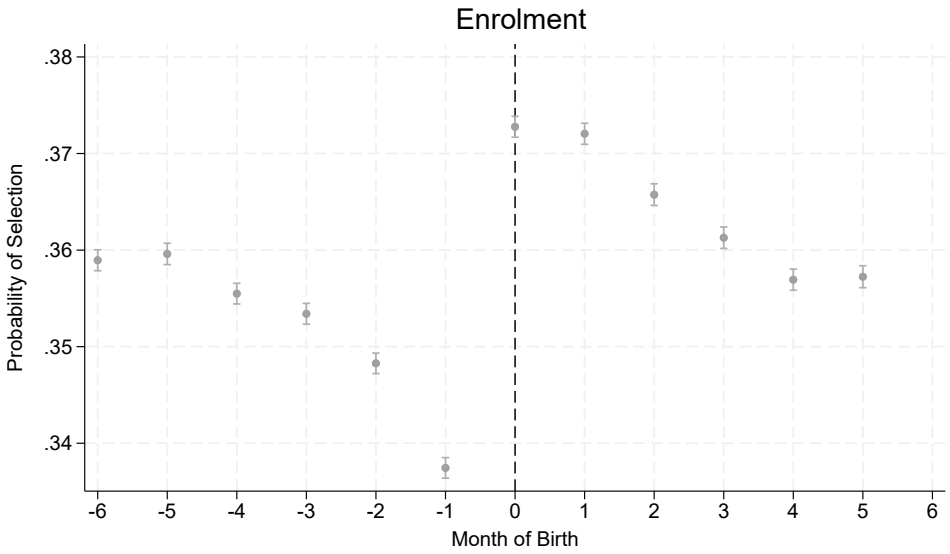


Figure A.3: Probability of university enrolment by month of birth, centred at 0 for September-born students, illustrating the discontinuity around the academic year cutoff.

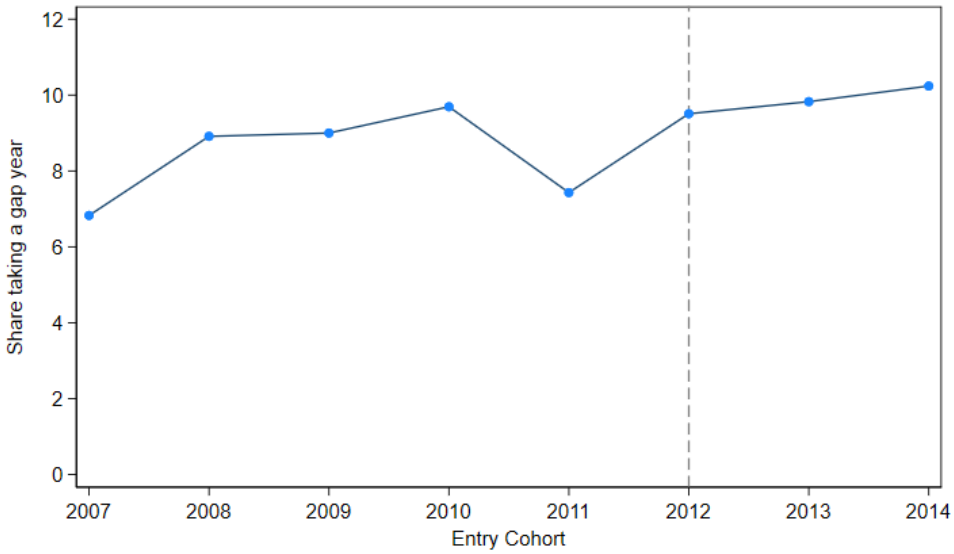


Figure A.4: Share of Students Delaying University Entry by One Year

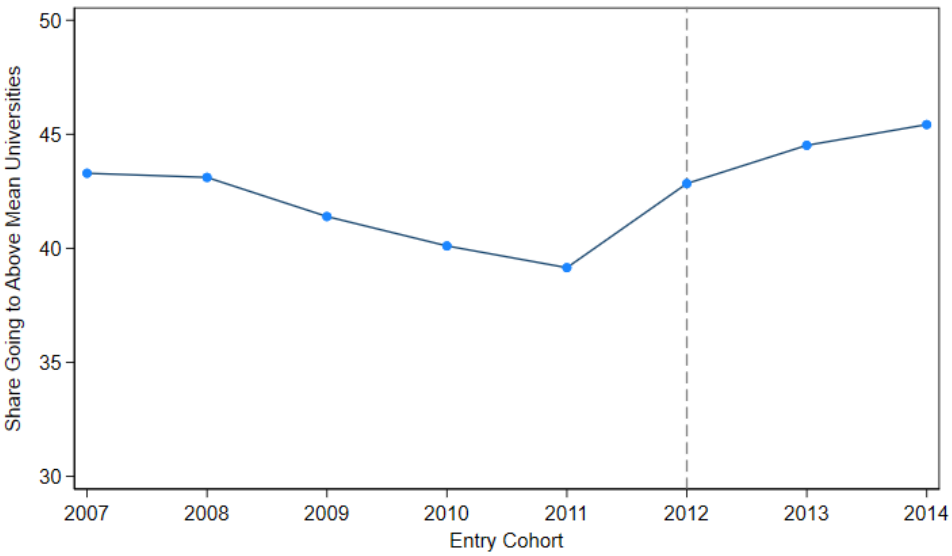


Figure A.5: Share of Entrants Attending Selective Institutions Over Time

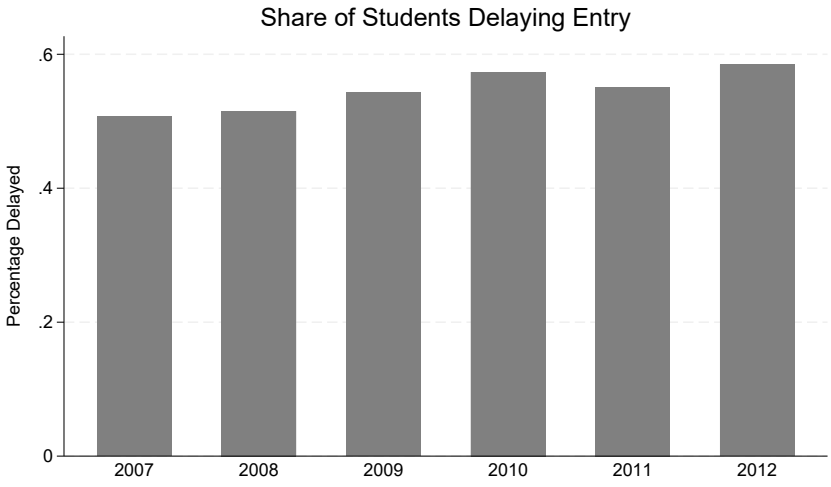


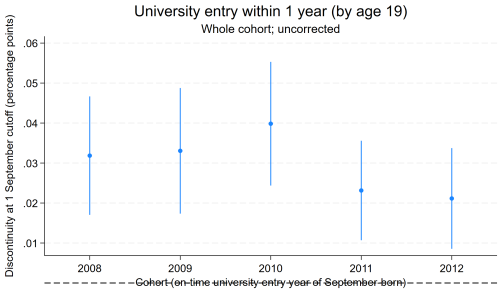
Figure A.6: Share of students who delay school entry by one year, by cohort.

Table A.1: Covariate Balance at the 1 September Cutoff, by Cohort

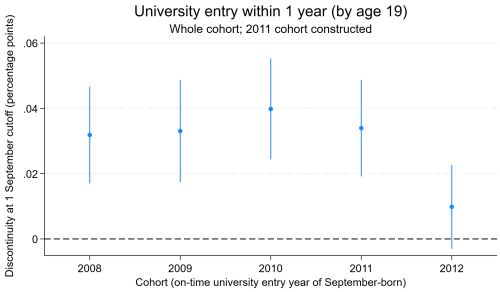
	(1) 2008	(2) 2009	(3) 2010	(4) 2011	(5) 2012
Female	-0.4	-0.3	-0.7	-0.5	-1.0
	$[-1.5,\ 0.6]$	$[-1.7,\ 1.1]$	$[-2.1,\ 0.7]$	$[-2.0,\ 1.1]$	$[-2.5,\ 0.6]$
N	337,105†	252,295	249,650	244,150	242,865
Eligible for FSM	-2.9^{*}	-0.8	-1.1^{*}	-0.7	-0.2
	$[-4.1,\ -1.7]$	$[-2.0,\ 0.3]$	$[-2.0,\ -0.2]$	$[-1.7,\ 0.3]$	$[-1.3,\ 0.9]$
N	240,180	252,295	249,650	244,150	242,865
Low-participation area	-0.2	0.7	0.3	-1.5^{*}	0.8
	$[-1.4,\ 1.1]$	$[-0.6,\ 1.9]$	$[-1.1,\ 1.7]$	$[-2.9,\ -0.1]$	$[-0.4,\ 2.0]$
N	240,180	252,295	249,650	244,150	338,325†

Notes: Each cell reports the regression discontinuity estimate at the cutoff indexed by the column year, in percentage points, with the reported 95 percent confidence interval in brackets. * indicates that the reported interval excludes zero. Counts rounded to the nearest five.

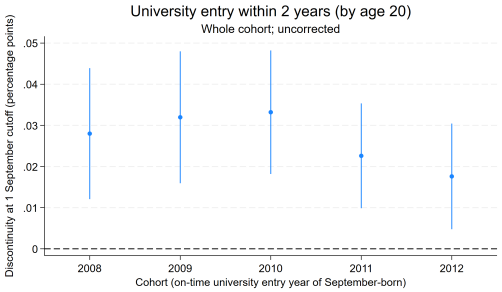
Figure A.7: Cumulative University Entry at Intermediate Horizons: Annual Discontinuities



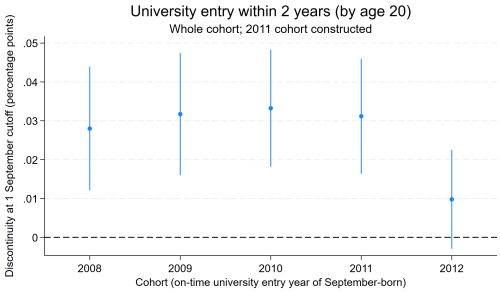
(a) Within 1 year, original



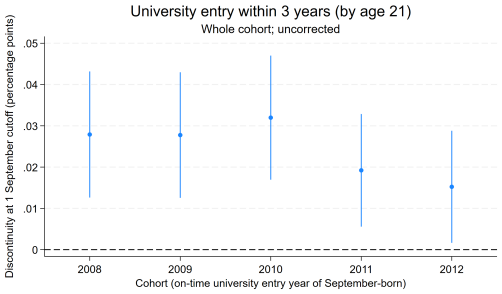
(b) Within 1 year, constructed



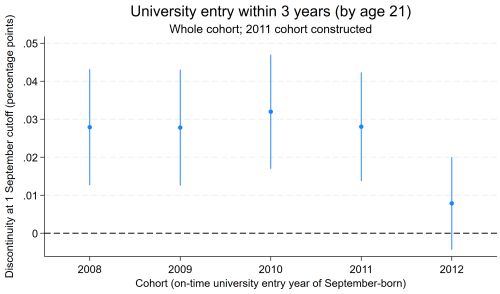
(c) Within 2 years, original



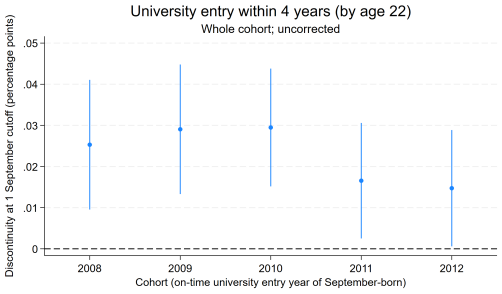
(d) Within 2 years, constructed



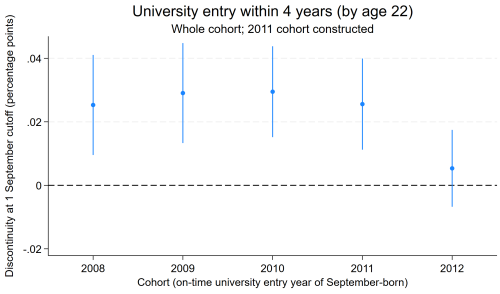
(e) Within 3 years, original



(f) Within 3 years, constructed



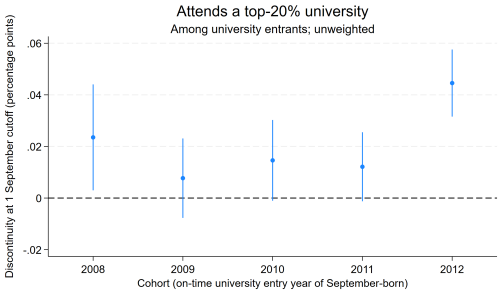
(g) Within 4 years, original



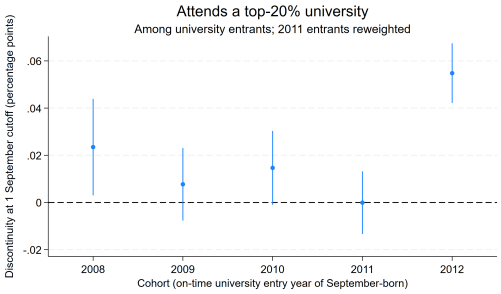
(h) Within 4 years, constructed

Note: Outcomes are indicators for entering university within one to four years of the on-time date, defined for the whole cohort. Left panels use observed outcomes; right panels use outcomes for the 2011 entry cohort constructed from the entry process of earlier cohorts

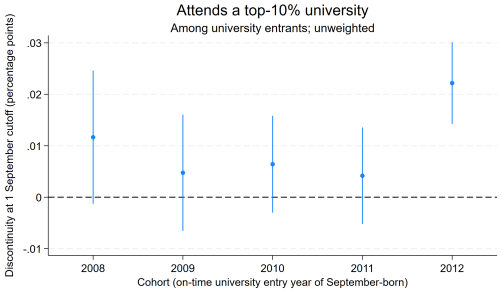
Figure A.8: Selectivity at Higher Thresholds: Annual Discontinuities



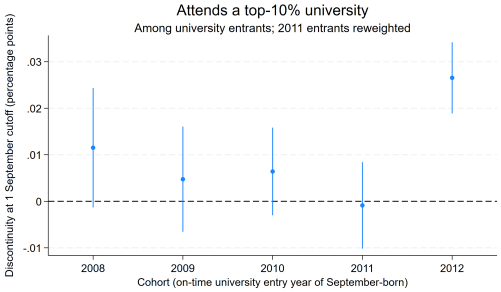
(a) Top 20%, conditional, unweighted



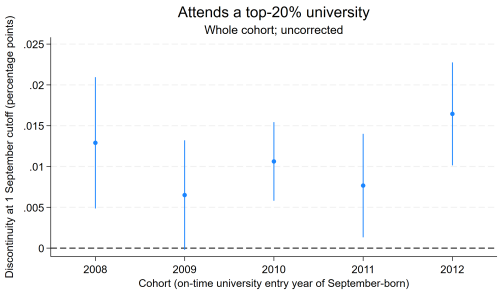
(b) Top 20%, conditional, reweighted



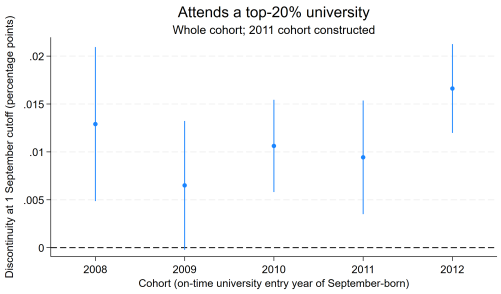
(c) Top 10%, conditional, unweighted



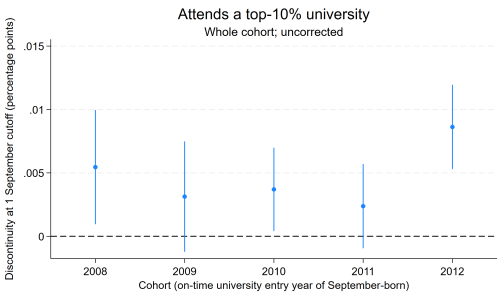
(d) Top 10%, conditional, reweighted



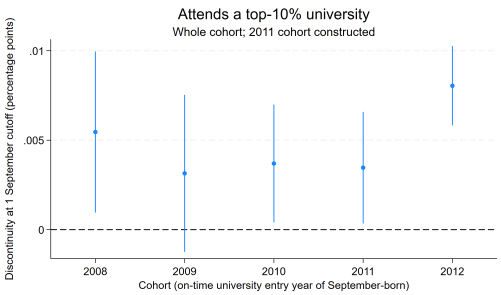
(e) Top 20%, whole cohort, original



(f) Top 20%, whole cohort, constructed



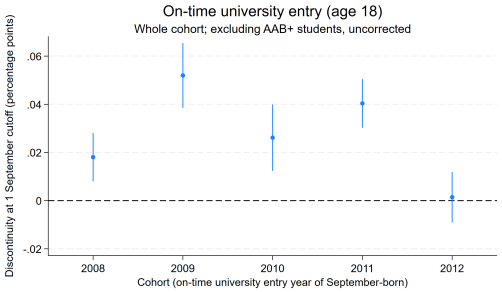
(g) Top 10%, whole cohort, original



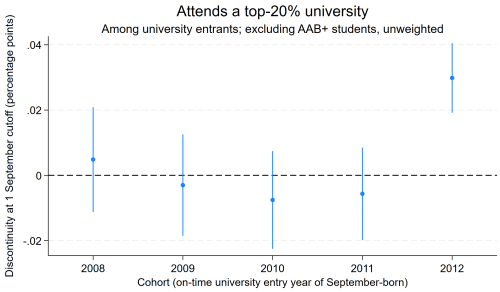
(h) Top 10%, whole cohort, constructed

Note: Panels (a)–(d) are defined among university entrants, with the 2011 entrants reweighted in (b) and (d). Panels (e)–(h) are defined for the whole cohort, with the 2011 cohort's outcomes constructed in (f) and (h).

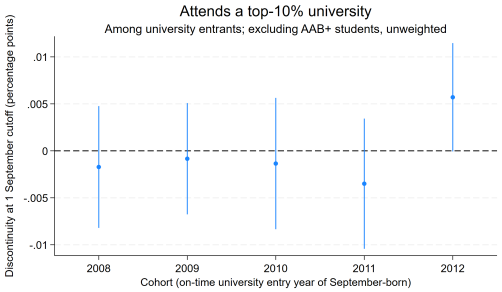
Figure A.9: Excluding AAB+ Students, Additional Outcomes: Annual Discontinuities



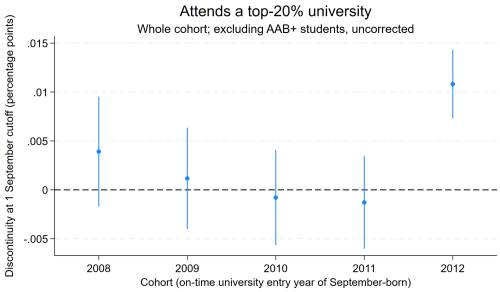
(a) On-time entry



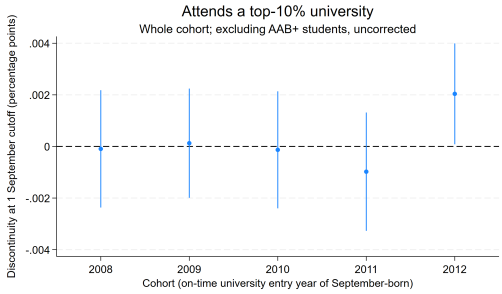
(b) Top 20%, conditional



(c) Top 10%, conditional



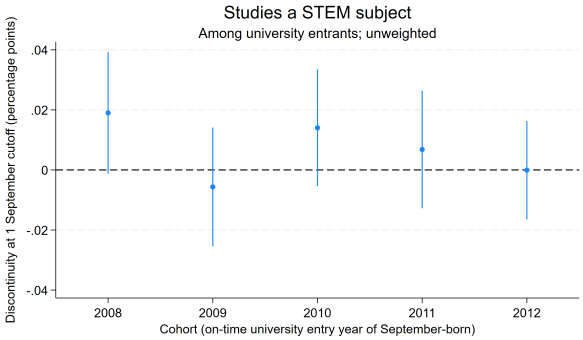
(d) Top 20%, whole cohort



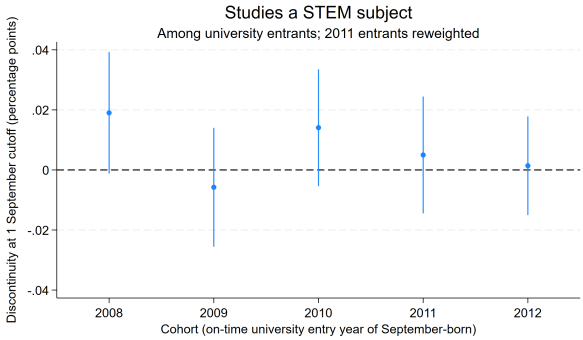
(e) Top 10%, whole cohort

Note: The sample excludes students achieving AAB or higher at A-level. All panels use observed, unweighted outcomes. Pooled estimates for the whole-cohort outcomes are pending.

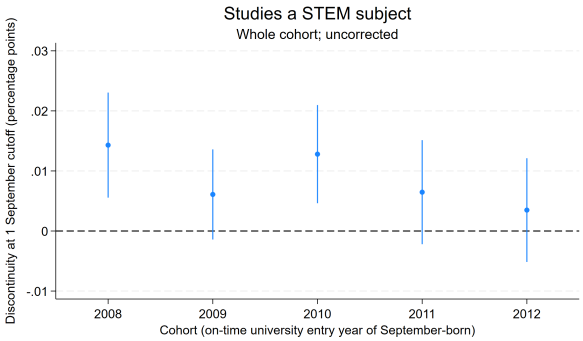
Figure A.10: STEM Participation: Annual Discontinuities



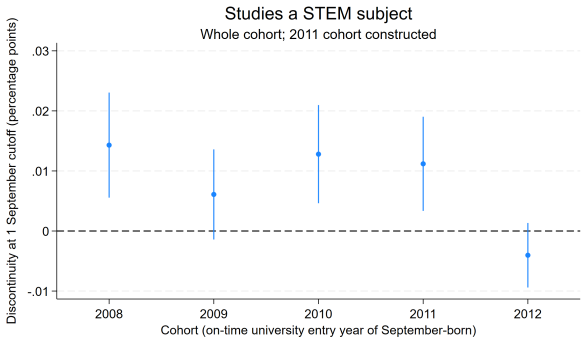
(a) STEM among entrants, unweighted



(b) STEM among entrants, reweighted



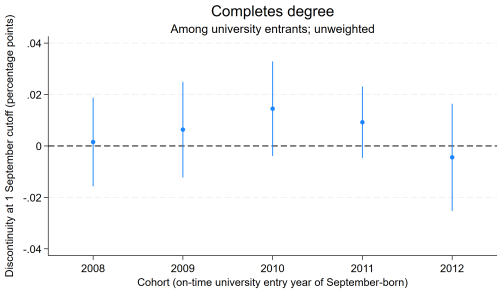
(c) STEM, whole cohort, original



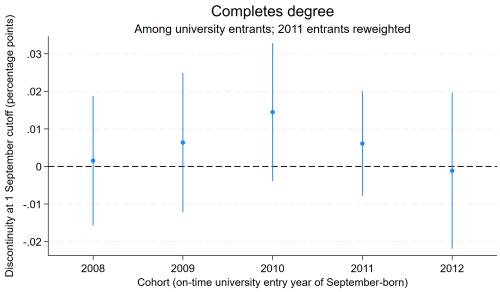
(d) STEM, whole cohort, constructed

Note: Panels (a) and (b) are defined among university entrants; panel (b) reweights the 2011 entrants. Panels (c) and (d) are defined for the whole cohort; panel (d) constructs the 2011 cohort's outcomes.

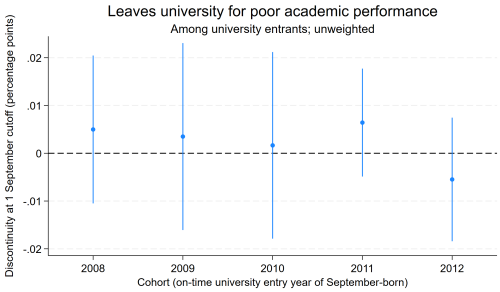
Figure A.11: Completion, Withdrawal, and Postgraduate Study: Annual Discontinuities



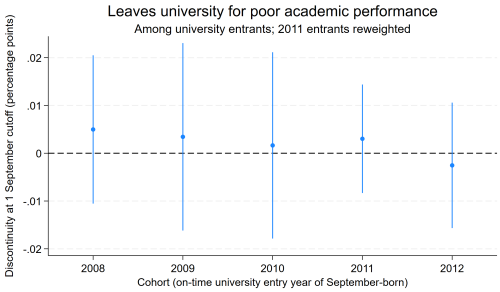
(a) Degree completion, unweighted



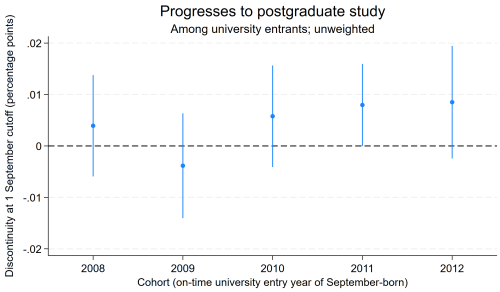
(b) Degree completion, reweighted



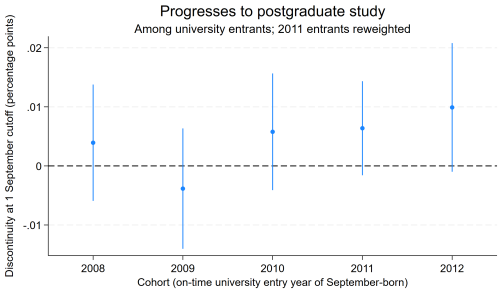
(c) Withdrawal (poor academic performance), unweighted



(d) Withdrawal (poor academic performance), reweighted



(e) Postgraduate study, unweighted



(f) Postgraduate study, reweighted

Note: All outcomes are defined among university entrants; right panels reweight the 2011 entrants.